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Abstract

This research focuses on developing machine learning-based predictive models for heart disease diagnosis using the UCI Heart Disease dataset. The study encompasses comprehensive data preprocessing, exploratory data analysis (EDA), and feature engineering to enhance the dataset's predictive potential. Various machine learning algorithms, including Decision Trees, Random Forest, SVM, and Neural Networks, are evaluated with a focus on model selection and hyperparameter tuning.

The research emphasizes the importance of K-Fold Cross-Validation and Grid Search techniques in optimizing model performance. The study aims to contribute to the advancement of accurate and reliable heart disease diagnosis, utilizing a dataset renowned for its extensive cardiovascular health information. The findings offer insights into algorithm performance, laying the groundwork for the development of effective diagnostic tools with the potential to impact cardiovascular health outcomes positively.

Acronyms

S. N	Acronyms	Full form
1	AGI	Artificial General Intelligence
2	NLP	Natural Language Processing
3	LLM	Large Language Model
4	DM	Diffusion Models
5	AUC	Area Under the Curve
6	ROC	Receiver Operating Characteristic Curve
7	ANOVA	Analysis of Variance
8	KNN	k-nearest neighbors
9	PCA	Principle Component Analysis
10	MSE	Mean Squared Error
11	SVM	Support Vector Machine
12	SWOT	Strengths, Weaknesses, Opportunities and Threats
13	NLP	Neuro-linguistic programming (psychology)
14	CRISP DM	**CR**oss **I**ndustry **S**tandard **P**rocess for **D**ata **M**ining
15	PCA	Principal Component Analysis
16	OLS Method	Ordinary Least Square Method

Glossaries

1) **Heuristics –** method of problem solving
2) andigha riche – a conclusion that provides more balanced picture, in which skill and heuristics are alternative sources of intuitive judgments and choices.
3) Plot Synopsis
4) Back propagation - An algorithm called backpropagation, often known as backward propagation of mistakes, is made to check for faults by looping backward from output nodes to input nodes. It is a crucial mathematical technique for enhancing machine learning and data mining prediction accuracy.
5) Ordinary Least Square (OLS) Method – Method to calculate the coefficients of Multiple Linear Regression using Matrix & Linear Algebra

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1) Introduction

Machine intelligence is the last invention that humanity will ever make. – Nick Bostrom

Astrology is the longest running data science project in human history. - Unknown

Heart disease is one of the most significant health challenges in the world. It as the leading cause of global deaths. Heart disease includes different cardiovascular diseases (CVD) that affect the heart and blood vessels, posing a serious threat to people around the world. World Health Organization (WHO) reports show a worrying trend, estimating that almost 23.6 million people will have CVD by 2030. This shows the urgent need to understand the complexities of heart diseases.

In the United States, CVD is a major health concern, causing 1 in 4 deaths. This highlights how much heart diseases impact people's health globally. The Middle East and North Africa (MENA) region face an even higher percentage, with CVD causing 39.2% of deaths. This makes it important to closely look at what contributes to heart diseases and find ways to deal with their impact.

Detecting heart issues early and having effective treatments are crucial to reducing deaths from cardiovascular diseases. It's especially important for people at high risk of developing heart disease. Our exploration into heart diseases aims to look at different aspects, using information from WHO. This approach seeks to provide a detailed understanding of risk factors, how to diagnose, and ways to treat, all based on evidence. (World Health Organization (WHO), 2021)

In this project, the heart disease dataset from the University of California, Irwing has been used. The dataset comprises various attributes or features that provide information about an individual's health. These features include both demographic and medical characteristics. Common attributes include:

Attributes

- **Age**: Age of the patient.
- **Sex**: Gender of the patient (1 = male, 0 = female).
- **Chest Pain Type**: Type of chest pain experienced by the patient.
- **Resting Blood Pressure**: Resting blood pressure in mm Hg.
- **Cholesterol Levels**: Serum cholesterol in mg/dl.
- **Fasting Blood Sugar**: Fasting blood sugar > 120 mg/dl (1 = true; 0 = false).
- **Resting ECG**: Resting electrocardiographic results.

- **Max Heart Rate**: Maximum heart rate achieved.
- **Exercise-Induced Angina**: Presence of angina induced by exercise (1 = yes; 0 = no).
- **Oldpeak**: ST depression induced by exercise relative to rest.
- **ST_Slope (Number of Major Vessels)**: Number of major vessels colored by fluoroscopy.

Target Variable: (HeartDisease)

This dataset will be utilized for binary classification, where the goal is to predict the presence or absence of heart disease. The target variable typically takes the value 0 if no heart disease is present and 1 if heart disease is present.

Size:

The dataset contains 918 instances (rows) and 12 columns.

Role of Machine Learning in Heart Disease Prediction

Machine learning models play a crucial role in predicting heart disease by examining diverse factors related to an individual's health. These models analyze information like medical history, lifestyle choices, and genetic data to assess the risk of developing heart-related issues. By considering multiple factors simultaneously, these models provide a comprehensive evaluation, helping healthcare professionals identify individuals at a higher risk early on and enabling proactive measures.

Moreover, machine learning algorithms are adept at early detection of heart disease. They examine medical imaging data, such as X-rays, MRI scans, and echocardiograms, to identify subtle abnormalities that may indicate the initial stages of heart problems. Early detection is vital for timely interventions and better management of the condition, potentially preventing further complications.

Machine learning models personalize predictions by tailoring them to an individual's unique characteristics. Factors like age, gender, blood pressure, cholesterol levels, and other relevant health metrics are considered, resulting in more accurate and individualized risk assessments. This personalized approach enhances the precision of predictions and allows for targeted preventive strategies. (Cuocolo R, Perillo T, De Rosa E, Ugga L, Petretta M., 2019)

Additionally, these models integrate data from various sources, such as electronic health records and wearable devices. This holistic approach provides a comprehensive understanding of an individual's health profile, contributing to more informed predictions and better decision-making by healthcare professionals.

Wearable devices equipped with machine learning capabilities enable continuous monitoring of vital signs, facilitating real-time data collection. This remote monitoring allows for the early detection of potential issues and timely interventions, enhancing the overall management of heart health. (Sabry F, Eltaras T, Labda W, Alzoubi K, & Malluhi Q, 2022 April)

The continuous validation and refinement of machine learning models using new data contribute to improved accuracy and reliability over time. This iterative process ensures that the models adapt to evolving patterns and factors influencing heart disease, making them more effective in predicting and preventing cardiovascular issues.

In summary, machine learning models in heart disease prediction offer a multifaceted approach, combining risk assessment, early detection, personalized predictions, and remote monitoring. These models empower healthcare professionals with valuable insights, ultimately contributing to more effective strategies for preventing and managing heart-related conditions.

1.1 Objectives

The main objective of my dissertation is to address the significant health challenges posed by heart disease through the development and evaluation of advanced machine learning models for accurate prediction. The objectives address various facets, each contributing to the goal of enhancing cardiovascular risk assessments and improving patient outcomes.

Literature Review:

- Explore heart disease prediction landscape through a comprehensive literature review.
- Identify limitations of traditional risk assessment methods.
- Review current machine learning applications.

Data Collection:

- Collect diverse datasets including health records, imaging, lifestyle, and genetic data.
- Integrate data for a robust dataset for model development.

Model Development:

- Develop machine learning models using algorithms like Decision Trees, Random Forest, KNN (K-nearest Neighbors) and SVM (Support Vector Machine).
- Evaluate models rigorously with established metrics.

Explainability and Features:

- Understand which features matter and make the model easy to interpret.
- Prioritize influential features in heart disease prediction like feature selection.

Comparative Analysis:

- Compare machine learning models with traditional risk assessment methods.
- Validate against independent datasets and real-world scenarios.

Ethical Considerations:

- Address biases, privacy concerns, and ethical implications in heart disease prediction.
- Explore challenges in real-world implementation in healthcare settings.

Conclusion:

- Contribute to improving cardiovascular risk assessment and patient care.
- Develop accurate, interpretable, and ethically sound machine learning models.

1.2 Data Understanding

S. N	Features	Data Type
1	Age (years)	int64
2	Sex 1: male 0: female	object
3	ChestPainType TA - Typical Angina ATA - Atypical Angina NAP - non-Anginal Pain ASY - Asymptomatic	object
4	RestingBP	int64
5	Cholesterol	int64
6	FastingBS	int64
7	RestingECG	object
8	MaxHR	int64
9	ExerciseAngina	object
10	Oldpeak	Float64
11	ST_Slope	object
12	HeartDisease 1: True 0: False	int64

Table 1: Data Understanding table

2) Literature Review

In the literature review section, various similar research studies have been illustrated, incorporating findings from books, research papers, journal articles, and Google Scholar. The exploration commenced with the introduction of machine learning concepts, techniques, and models utilized in the prediction process. The shortcomings identified in these researches were outlined, accompanied by proposed improvements for addressing these limitations.

2.1 Machine Learning

Machine learning is a significant area of artificial intelligence that is leading the way in technological advancement and gives computers the capacity to learn from data and adapt in ways similar to how the human mind does. The creation of algorithms that enable computers to recognize patterns and reach well-informed conclusions without needing to be specifically programmed for every situation is the core of machine learning. New opportunities in data analysis, automation, and problem solving have been made possible by this paradigm change in computing.

One of the key principles underlying machine learning is the capacity to recognize patterns within datasets. By exposing algorithms to ample examples, these systems become adept at generalizing and extrapolating information. For instance, a machine learning model can be trained on extensive datasets of handwritten digits, enabling it to accurately recognize and classify new digits it has never encountered before. (Baheti, 2022)

Machine learning encompasses a variety of techniques, each designed for specific tasks and types of data. Some of the common machine learning techniques are:

1) Supervised Learning:

Supervised learning is an approach in machine learning, where models are trained on labeled datasets. In this context, the algorithm learns the relationship between input data and corresponding output labels. For instance, in a supervised learning scenario, the algorithm could learn to differentiate between spam and non-spam emails based on labeled examples. Accurately predicting the output labels for newly discovered, unobserved data is the aim of learning a mapping or function.

Examples:

- **Regression**: Predicting a continuous output, such as house prices based on features like square footage and location.

- **Classification**: Assigning labels to input data (e.g., spam detection, image recognition).

2) Unsupervised Learning:

Contrastingly, in unsupervised learning, models are trained using unlabeled data, aiming to identify existing patterns and structures. This can be likened to exploring a collection of books without knowing their genres, as the algorithm autonomously discovers similarities and groupings within the data. (Delua, 2021)

Examples:

- **Clustering**: Grouping similar data points together, such as identifying customer segments based on purchasing behavior.

- **Dimensionality Reduction**: Reducing the number of features while retaining important information, e.g., using Principal Component Analysis (PCA).

3) Semi-Supervised Learning:

Semi-supervised learning is a combination of both supervised and unsupervised learning. The algorithm is trained on a dataset that contains both labeled and unlabeled data. This method works well in situations where getting tagged data is costly or time-consuming.

Example:

Training a model on a dataset with limited labeled examples for fraud detection, supplemented with a larger amount of unlabeled data.

4) Reinforcement Learning:

Reinforcement learning introduces an interactive dimension to machine learning, where an agent learns by receiving feedback in the form of rewards or penalties based on its actions. This dynamic approach is akin to a computer learning to play a game through trial and error, adjusting its strategy based on the outcomes of its actions.

Example:

Training a computer program to play games like chess or Go, where the agent learns from trial and error.

5) Deep Learning:

Deep learning is a subset of machine learning that focuses on neural networks with multiple layers, known as deep neural networks. These networks automatically learn hierarchical features from data and are particularly effective in capturing complex patterns.

Examples:

- **Convolutional Neural Networks (CNNs)**: Used for image recognition tasks.

- **Recurrent Neural Networks (RNNs)**: Suitable for sequential data, like time series or natural language.

6) **Ensemble Learning:**

Ensemble learning enhances overall performance and generalization by combining predictions from several models. In order to produce a better, more reliable model, it makes use of the diversity of individual models.

Examples:

- **Random Forests**: An ensemble of decision trees that collectively make predictions.

- **Gradient Boosting**: Builds a strong model by combining weak models sequentially.

7) **Transfer Learning:**

Transfer learning involves pre-training a model on a large dataset and then fine-tuning it for a specific task with a smaller dataset. This approach is especially useful when the target task has limited labeled data.

Example:

Using a pre-trained neural network for image recognition and adapting it for a specific set of images or classes.

8) **Instance-Based Learning:**

Instance-based learning methods make predictions based on the similarity of new instances to instances in the training set. These methods don't explicitly learn a model but rely on stored instances for prediction.

Example:

- **k-Nearest Neighbors (k-NN)** algorithm, which predicts the label of a new instance based on the labels of its k-nearest neighbors in the training set.

Each of these machine learning techniques has its own strengths and weaknesses, and the choice of which to use depends on the nature of the data, the problem to be solved, and the goals of the modeling task. (Bishop, 2006)

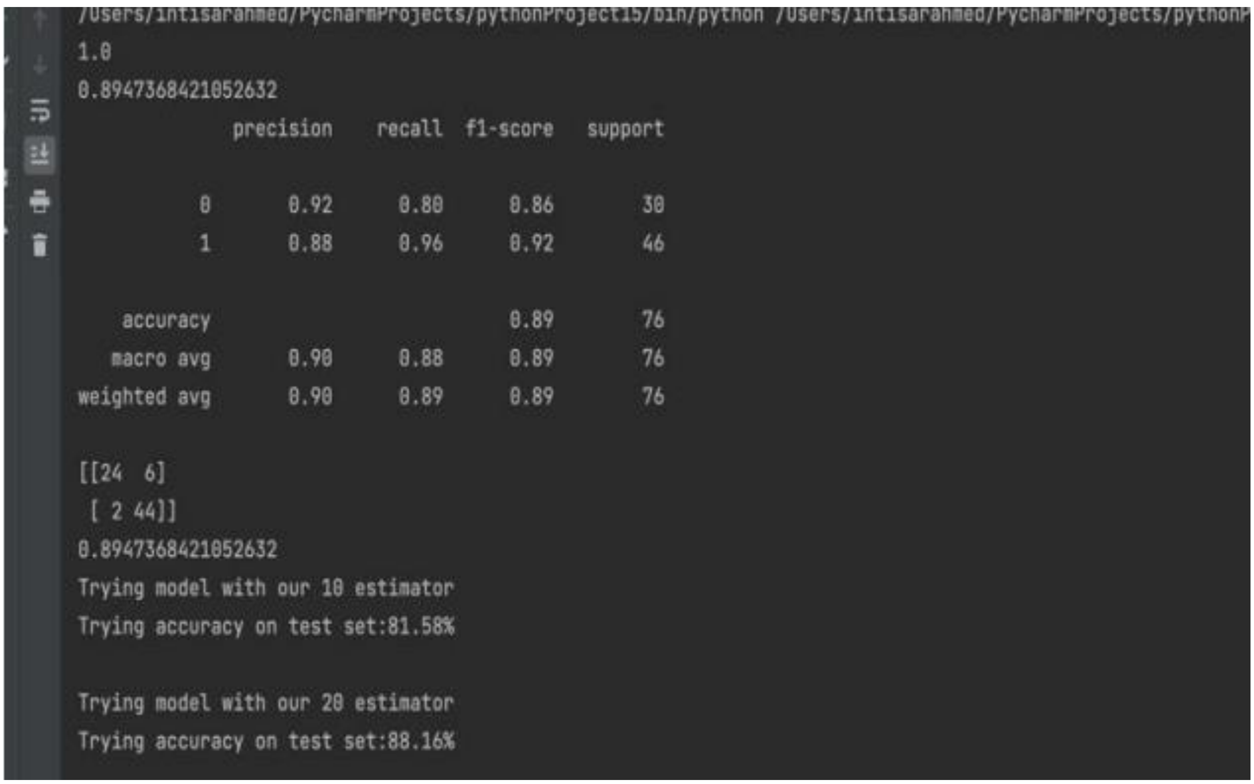
The applications of machine learning are extensive, spanning diverse domains such as image recognition, natural language processing, healthcare, finance, and more. Its impact is evident in the automation of complex tasks, the development of predictive models, and the extraction of meaningful insights from vast and intricate datasets. (Elahi, Afolaranmi, Lastra , & Garcia , 2023)

Machine learning has the unique ability to make sense of data and improve over time, and thus continues to shape the technological landscape. This will pave the path for further progress of machine learning tasks that are further advanced and game changing.

2.2 Lists of Researches Integrated

1) First Project Research

Intisar Ahmed, a graduate student from California State University, San Bernardino, has done his dissertations on topic “A Study of Heart Disease diagnosis using Machine Learning and Data Mining”. His dissertations dates back to December of 2022. He has employed Random Forest algorithm in his research. He has trained and tested the model. He went further to check accuracy by changing hyperparameter values. As a result, he was able to predict the heart disease with 89.4% accuracy and recall of 0.96. His results are shown below. (Ahmed, 2022)



```
/Users/intisarahmed/PycharmProjects/pythonProject15/bin/python /Users/intisarahmed/PycharmProjects/pythonP
1.0
0.8947368421052632
              precision    recall  f1-score   support

           0       0.92      0.80      0.86        30
           1       0.88      0.96      0.92        46

    accuracy                           0.89        76
   macro avg       0.90      0.88      0.89        76
weighted avg       0.90      0.89      0.89        76

[[24  6]
 [ 2 44]]
0.8947368421052632
Trying model with our 10 estimator
Trying accuracy on test set:81.58%

Trying model with our 20 estimator
Trying accuracy on test set:88.16%
```

Figure 1: Results on PyCharm using Random Forest Algorithm

Precision	Recall	F1-Score	Support
0.88	0.96	0.92	46

Figure 2: Confusion Matrix results

Probable Steps to take to improve this project

In this project, Cross Validation techniques such as K-Fold Cross-Validation, Stratified K-Fold Cross-Validation, Leave-One-Out Cross-Validation (LOOCV), etc., have not been employed by the researcher. The model's performance and its ability to generalize to new data can be assessed using Cross Validation techniques.

Cross Validation technique repeatedly splits the dataset into training and testing sets and offers a more robust performance estimate. It also helps to identify overfitting and facilitates hyperparameter tuning. It also aids in model selection, balances bias and variance in evaluations. Cross-validation makes machine learning models more reliable and generalizable.

Hence, Cross Validation techniques, preferably K-Fold Cross Validation, will be employed to improve the performance of the models.

2) Second Project Research

The second topic of research is “Heart Disease Prediction using Machine Learning Techniques”. This research paper was published in Wesleyan Journal of Research. The three authors of this research paper are from Baba Ghulam Shah Badshah University Rajouri, J&K, India (names are the source section). Their dissertations dates back to March 2021.They have employed SVM (Support Vector Machine), Decision Tree and Random Forest models to predict the heart disease.

They have implemented data pre-processing procedures and feature (column) selection using heatmap quite extensively. However, they have overlooked to evaluate algorithm using cross validation techniques. They also have not used algorithm optimization techniques like hyperparameter tuning. Nevertheless, they were able to achieve impressive accuracy of 99.39% using Random Forest and recall of 0.98. Their results are shown below. (Nissa, Jamwal, & Mohammad, 2021)

Algorithm	Accuracy
Support Vector Machine	84.93 %
Decision Tree	97.59 %
Random Forest	99.39 %

Figure 3: Accuracy comparison of 3 models

Classifier	Recall	Precision	Specificity	F1-Score
SVM	0.90	0.76	0.81	0.82
DT	0.95	1.0	1.0	0.97
RF	0.98	1.0	1.0	0.99

Figure 4: Evaluation Metrics for 3 models

Probable Steps to take to improve this project

In this project, the knowledge gained from the heatmap has been employed to choose the best features that are positively correlated. Those particular features have been used to evaluate the model. However, the Cross Validation technique and Search-Grid CV have not been used to optimize the model performance. Hence, these methods will be used to improve the overall performance.

3) Third Project Research

The third research article was published by a team of researchers under title Prediction of Heart Disease Using a Combination of Machine Learning and Deep Learning. Their research publication dates back to March 2021. They have done comparative analysis using machine learning algorithms and calculated evaluation metrics like confusion matrix, accuracy, precision, specificity, sensitivity and F1 score.

They seem to have issue of overfitting due to their smaller data size. They have normalized the data, removed outliers and were able to get the highest accuracy of

83.29% by using K-Nearest Neighbors among other machine learning models. However, they were able to significantly improve their accuracy (92.7%) by using Artificial Neural Network of 3 layers (128 units, 64 unitis and 32 units) respectively from the start. (Bharti, et al., 2021)

Classifiers	Accuracy (%)	Specificity	Sensitivity
Logistic regression	83.3	82.3	86.3
K neighbors	84.8	77.7	85.0
SVM	83.2	78.7	78.2
Random forest	80.3	78.7	78.2
Decision tree	82.3	78.9	78.5
DL	94.2	83.1	82.3

Figure 5: Comparative analysis of ML models

Probable Steps to take to improve this project

In this project, the researchers found themselves in a difficult spot due to the smaller data size, leading to the problem of overfitting. Additionally, Artificial Neural Networks, chosen by them, are vulnerable to smaller datasets, further exacerbating the issue of overfitting. Structured data, which is well-organized and follows a consistent format, is being used in this project. Traditional models like decision trees and SVM work well with structured data, as they can easily interpret and learn patterns from this organized format. Furthermore, traditional models can be easily used for cross-validation and hyper-tuning, unlike Artificial Neural Networks, which are complex and time-consuming to hyper-tune.

3. Theoretical Foundation

The theoretical foundations for the project are as follows:

3.1 My list of implemented Machine Learning Models

3.1.1 Decision Tree

A decision tree in machine learning is a predictive model that organizes information into a tree-like structure. The process involves recursively selecting the best features to make decisions at each internal node, leading to subsequent nodes or leaves representing predicted outcomes. The tree is constructed during training, and each decision is based on specific features, creating a hierarchical flowchart of decisions. Decision trees are known for their interpretability, as they provide a transparent and visual representation of the decision-making process. They are valuable for tasks where understanding the logic behind predictions is crucial. (Yadav, 2018)

Imagine it as a game of 20 Questions, but for data. The tree asks yes or no questions about the data, follows the answers, and finally gives a prediction. Each question narrows down the possibilities, and it keeps going until it reaches a prediction. The cool part is, it's like a visual map that helps us understand how the prediction was made. So, decision trees are like friendly detectives in the world of data, helping us make sense of information and make smart predictions.

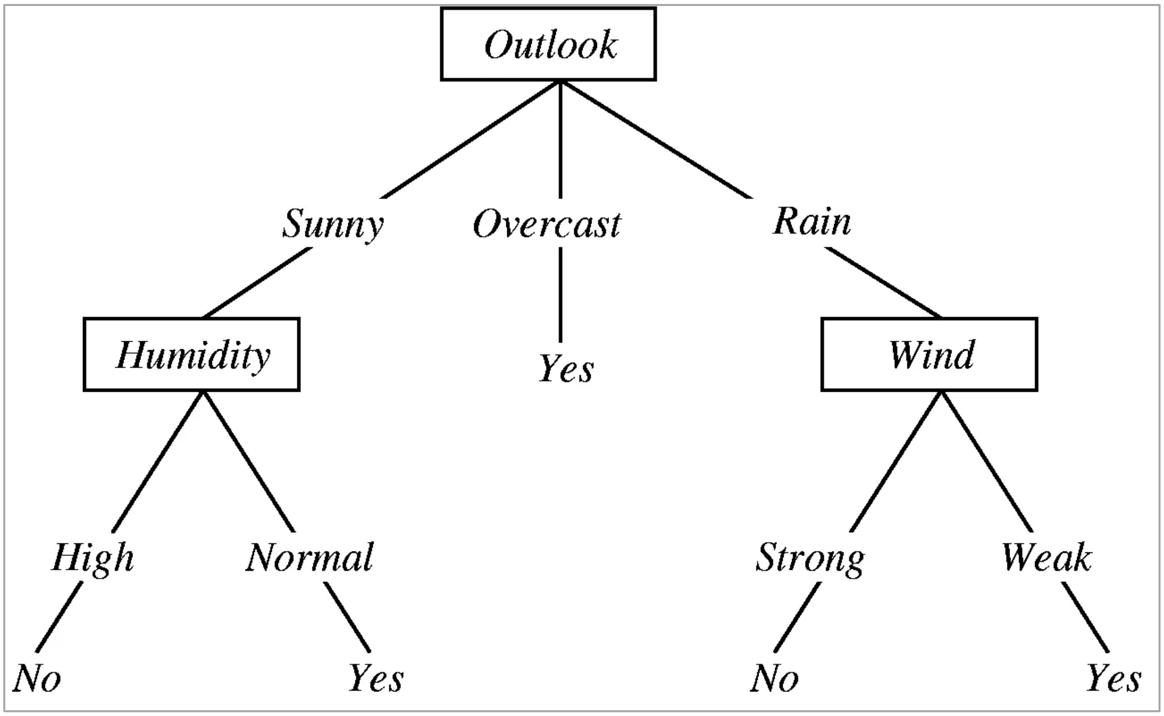
e 

Figure 6: Decision Tree for Rain Forecasting

e

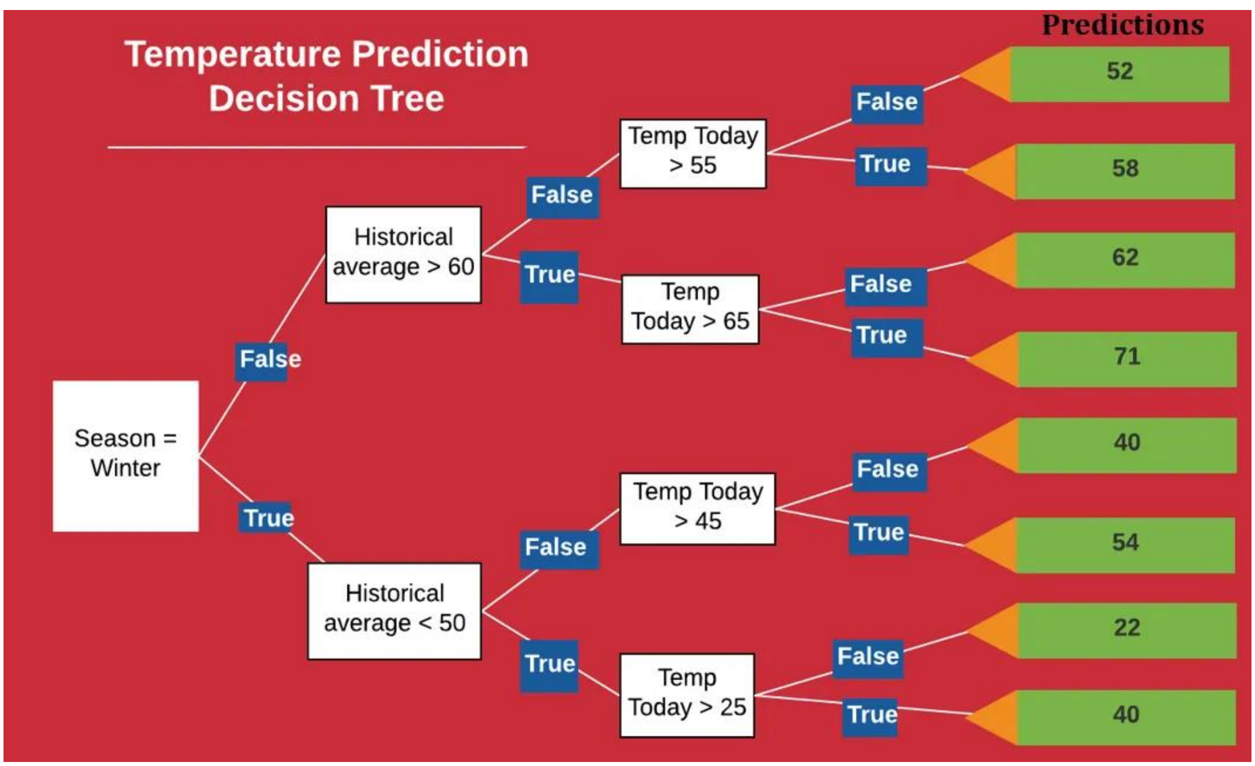


Figure 7L Machine Learning Decision Tree for Temperature Prediction

Reason for choosing

Decision Tress model is like transparent decision-makers, meaning they provide clear and understandable insights into how they reach conclusions. This is crucial for project, especially when predicting something as important as heart disease. It excels in handling the complex medical data, making them a reliable choice.

Additionally, decision trees are robust when faced with missing information, simplifying our data preparation process. When combined in an ensemble method like Random Forest, decision trees enhance their power and flexibility, making them a valuable asset for our project in predicting heart disease with clarity and accuracy. (Smolic, 2023)

3.1.2 Random Forest

Using the combined strength of several decision trees, Random Forest is an ensemble machine learning technique that produces predictions. The ultimate forecast is arrived at by combining the findings from every decision tree in the forest, each of which is autonomously built and produces a prediction. Utilizing a subset of characteristics and data points for every tree creates randomization while also increasing variety and decreasing overfitting.

This randomness helps to improve the model's robustness and reduces the risk of overfitting. The final prediction is then determined by aggregating the predictions of all the trees in the forest, resulting in a powerful and versatile algorithm known for its high accuracy and resilience against noise in the data. (E. R, 2024)

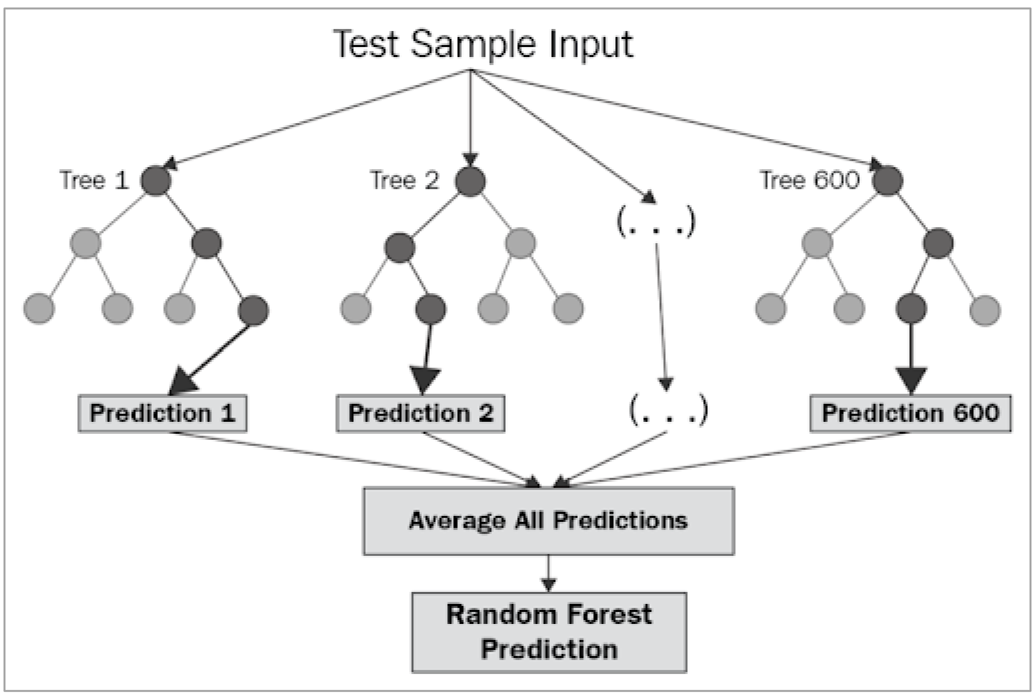


Figure 8: Decision Tree working mechanism

Reason for choosing

Random Forest is combines the strengths of multiple decision trees to make our predictions more accurate and robust. This teamwork helps overcome the limitations of individual trees, making Random Forest a powerful choice. Moreover, it's resilient to overfitting, ensuring our model doesn't get too obsessed with specific details. Therefore, Random Forest is a reliable, collaborative, and effective approach to predict heart disease, which makes it an excellent choice for the project.

3.1.3 Logistic Regression

Predicting the likelihood that an instance will belong to a specific class is the aim of binary classification problems using the supervised learning technique logistic regression. Logistic regression is used for classification, not regression, despite its name. Using the logistic function, it describes the link between the input features and the likelihood that an output will belong to a particular class.

The logistic function, also known as the sigmoid function, maps any real-valued number into the range of 0 to 1. During training, the algorithm adjusts its parameters to minimize the difference between predicted probabilities and actual class labels, typically using a method called maximum likelihood estimation. Logistic Regression is widely used due to its simplicity, interpretability, and efficiency, making it a foundational tool in the field of machine learning. (Brownlee, Logistic Regression for Machine Learning, 2023)

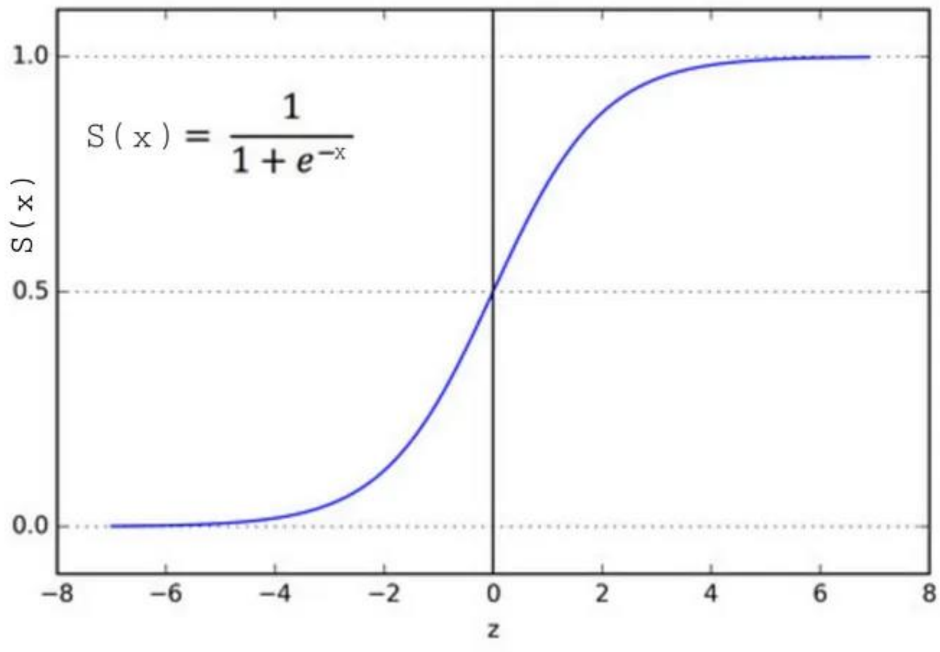


Figure 9: Logistic Regression

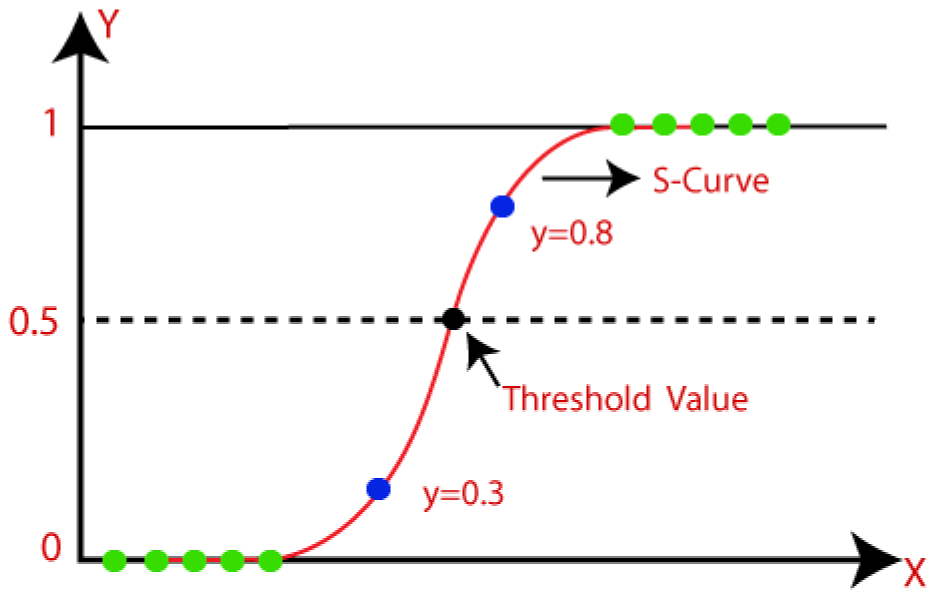


Figure 10: Logistic Regression

Reason for choosing

Logistic Regression excels in binary classification tasks, like predicting whether someone has heart disease or not. It's easy to interpret, making it perfect for explaining our predictions to healthcare professionals. Plus, logistic regression is computationally efficient, ensuring the model works quickly.

3.1.4 K-Nearest Neighbors (KNN)

K-Nearest Neighbors (KNN) is a widely used machine learning algorithm that serves both classification and regression purposes. Its principle is straightforward: for a given data point, KNN identifies the 'k' nearest neighbors in the feature space and predicts the outcome based on the majority class (for classification) or the average value (for regression) of these neighbors. This approach relies on the assumption that similar instances tend to share similar outcomes. (elastic, 2024)

In medical applications, such as disease prediction, KNN becomes valuable. For example, when diagnosing a patient, KNN assesses the class labels of nearby patients with similar features to predict the likelihood of a specific medical condition. The algorithm employs distance metrics, such as Euclidean distance, to quantify the closeness of data points in the multi-dimensional space.

One of the strengths of KNN lies in its simplicity and ease of interpretation. It operates without constructing an explicit model during training, making it particularly useful for scenarios where the underlying relationships are not well-defined or when immediate interpretability is crucial. However, its effectiveness can be influenced by the choice of the distance metric and the parameter 'k,' emphasizing the importance of thoughtful considerations in applying KNN to various contexts. (IBM, 2024)

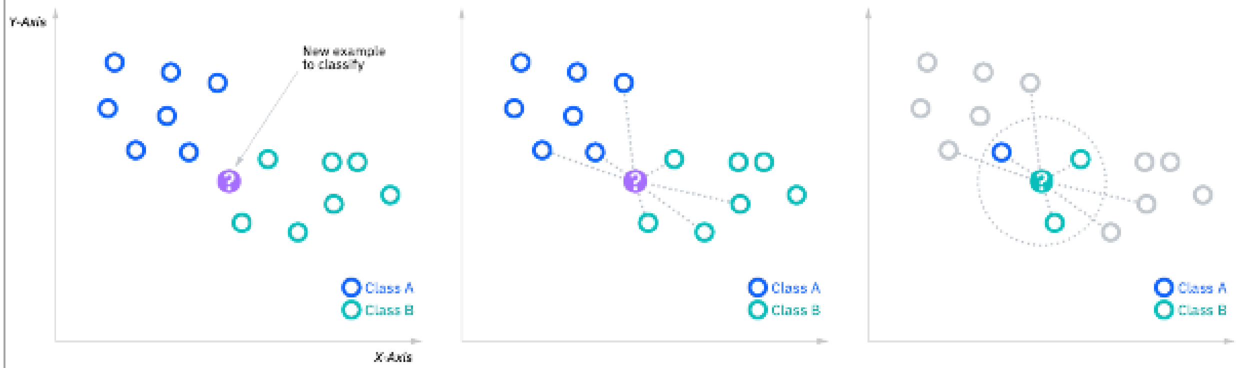


Figure 11: KNN diagram (IBM, 2024)

Reason for choosing

KNN is fantastic for classification tasks, like predicting heart disease, as it looks at the closest data points to make predictions. It's simple and doesn't make assumptions about the data, making it flexible for various situations. KNN is great at handling different patterns in the medical data, ensuring that the predictions are accurate.

4. Methodology

Here, qualitative research will be conducted, and conclusions will be drawn utilizing a numerical dataset. Numerical and statistical analyses will be performed to determine model evaluation metrics such as accuracy, precision, recall, F-1 score, etc.

1. Data Collection:

The key step in machine learning, data collection, involves the gathering of the right information to train the computer model. Relevant and accurate data is carefully selected to represent the problem to be solved. This step ensures that the model learns from quality information. Clear documentation is maintained to keep track of data sources. The stage for creating a strong and accurate machine learning model is set by successful data collection.

2. Exploratory Data Analysis (EDA):

Exploratory Data Analysis (EDA) is a critical phase in machine learning where the dataset is delved into to uncover important patterns and insights. During EDA, statistical and visual techniques are employed to understand the distribution of the data, identify trends, and spot any anomalies. Informed decisions about data preprocessing and feature engineering are facilitated by this process. By visualizing the data, a deeper understanding of its characteristics is gained, enabling better choices as the machine learning model is built and trained.

The characteristics of the data were brainstormed and understood through the performance of statistical analysis, visualizations, and summaries. The data was analyzed to identify patterns, correlations, outliers, and missing values.

3. Feature Engineering:

Feature engineering is a strategic process in a machine learning project where existing data is transformed or new features are created to enhance the performance of the model. The aim is to highlight important information and improve the model's ability to make accurate predictions by selecting or modifying features. Domain knowledge and creativity are often required in this step to extract valuable insights from the data. More robust and efficient machine learning models can be achieved through effective feature engineering, contributing significantly to the overall success of the project.

New features were created, existing features were transformed, and relevant features were selected to improve model performance. Features were normalized, scaled, and encoded as required by the model.

4. Model Selection:

Model selection is a crucial step in a machine learning project, where the most suitable algorithm for a specific problem is chosen. This decision is based on factors such as the nature of the data, the problem type (classification, regression, etc.), and the desired outcomes. Optimal performance and generalization on new, unseen data are aimed for by selecting the right model. The process involves evaluating different algorithms and picking the one that best aligns with the project's objectives. A thoughtful and informed model selection is essential for building a successful machine learning solution.

Appropriate machine learning algorithms or models were chosen based on the problem type and data characteristics. Different models such as Decision Trees, Random Forest, KNN, Naïve Bayes, etc., were experimented with, and their performance was evaluated using validation techniques like K-Fold cross-validation.

5. Model Training:

During this phase, patterns and relationships within the training data are learned by the selected algorithm. The model adjusts its internal parameters based on the provided examples, continuously improving its ability to make accurate predictions. Successful training relies on iterative optimization, where the model fine-tunes its parameters to minimize errors. The goal is to create a well-adjusted model that can generalize well to new, unseen data, allowing it to perform effectively in real-world scenarios.

The selected models were trained using the training dataset. Model parameters were then optimized using techniques like hyperparameter tuning, such as Search Grid cross-validation.

6. Model Evaluation:

Model evaluation is a critical step in assessing the performance of a trained machine learning model. During this phase, the model's predictions are compared to the actual outcomes on a separate set of data not used for training, often referred to as the test dataset. Various metrics, such as accuracy, precision, recall, and F1 score, are employed to measure how well the model generalizes to new, unseen data. A robust evaluation ensures that the model is effective and reliable in making predictions beyond the training set. This step is crucial for making informed decisions about the model's readiness for deployment and its potential impact on solving the intended problem.

The trained models were evaluated using appropriate evaluation metrics (accuracy, precision, recall, F1-score, etc.). Additionally, the models were validated using separate test datasets to assess generalization.

7. Documentation and Reporting:

Documentation involves recording key details about data, preprocessing, model specifics, and decisions made. This ensures transparency and facilitates collaboration. Project objectives, methodologies, and results are communicated to stakeholders clearly and concisely through reporting. These practices support transparency, reproducibility, and effective knowledge transfer within the project team and beyond.

In this section, clear and comprehensive documentation outlining the methodologies, findings, and models developed was maintained. In the report, the proposed approach, expected impact, and potential challenges were summarized.

5. Results/Findings

5.1 EDA (Exploratory Data Analysis)

Age Distribution in relation to Heart Disease

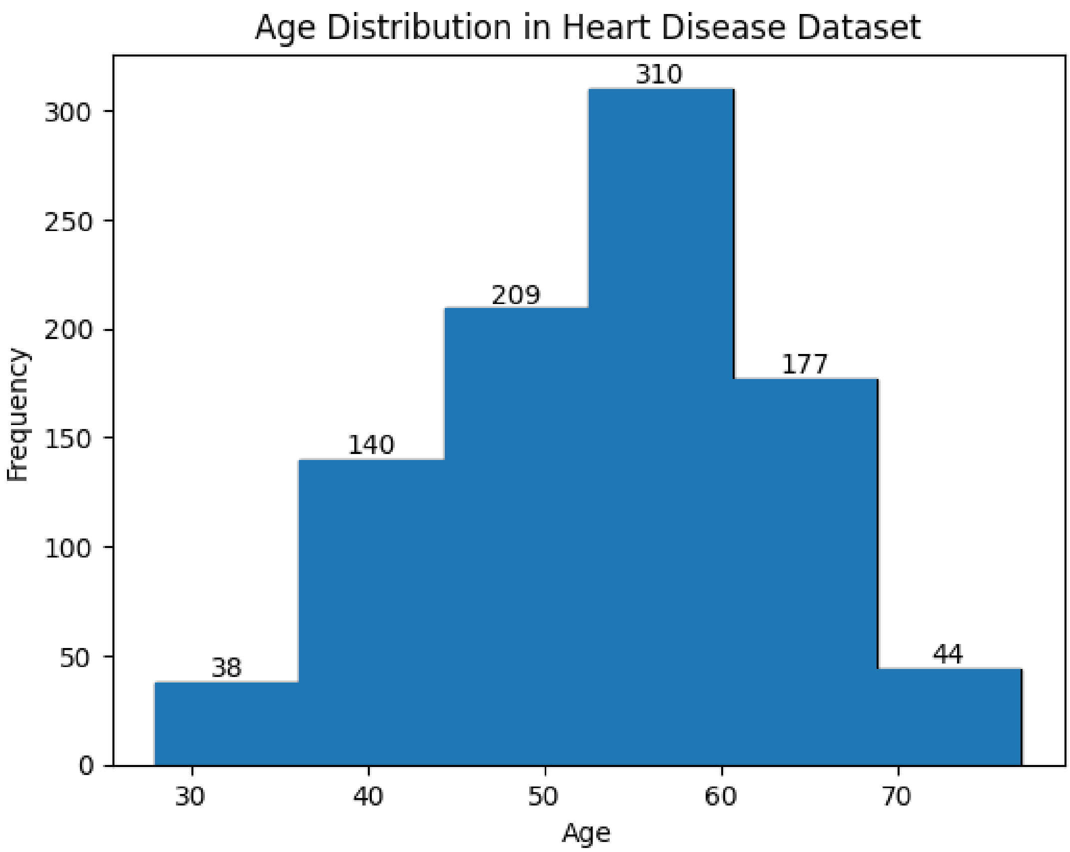


Figure 12: Age Distribution in Heart Disease

The above diagram shows the histogram distribution of age. Age group between about 53- 58 have the highest count of 310, followed by 46-51 with 209, followed by 58-63 with 177 and so forth. The value counts can be found on the top of the bins in the histogram.

Sex Ratio Distribution

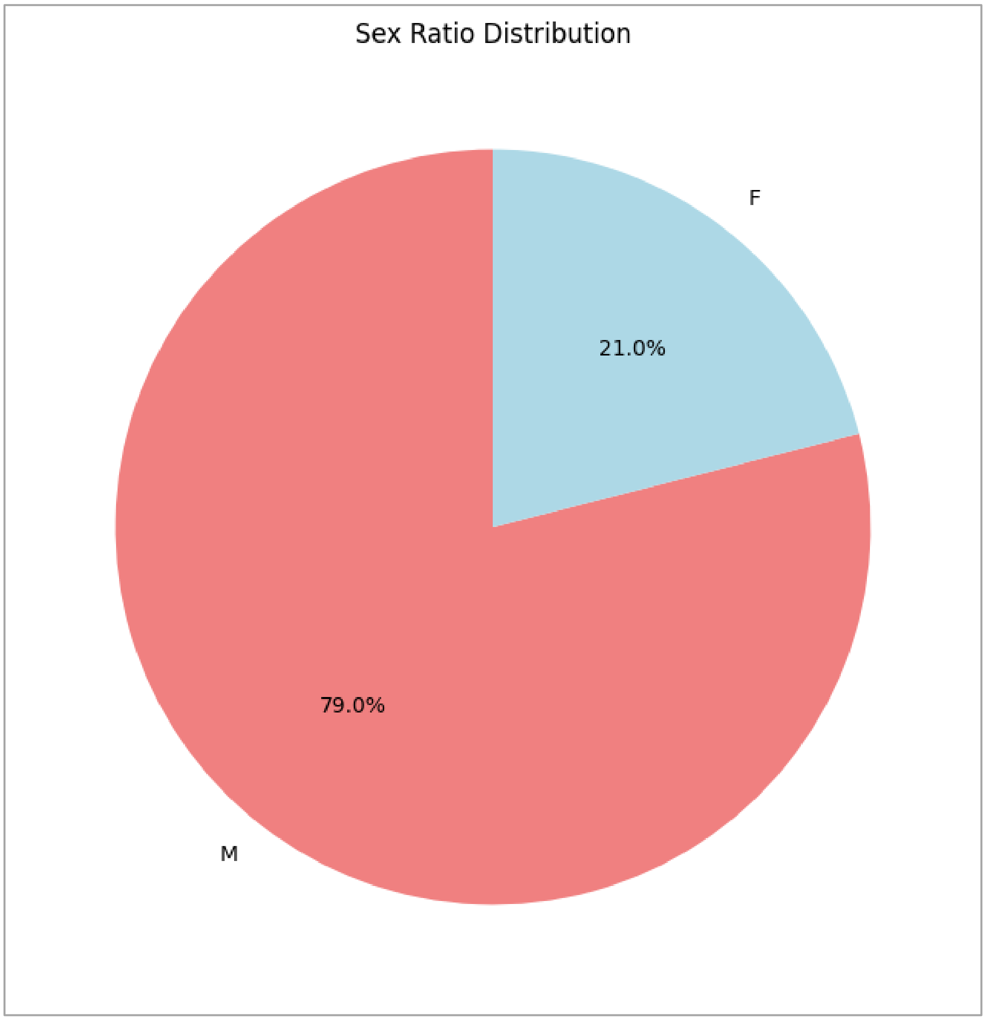


Figure 13: Pie chart showing sex ratio distribution

The above figure 13 shows the sex ratio of male and female on the dataset. The percentage of male is 79% while that of female is just 21%. The total numbers of male is 725 while the numbers of female is 193.

Histogram Distribution of Age with Heart Disease

Swarm Plot: Age and Heart Disease with Sex

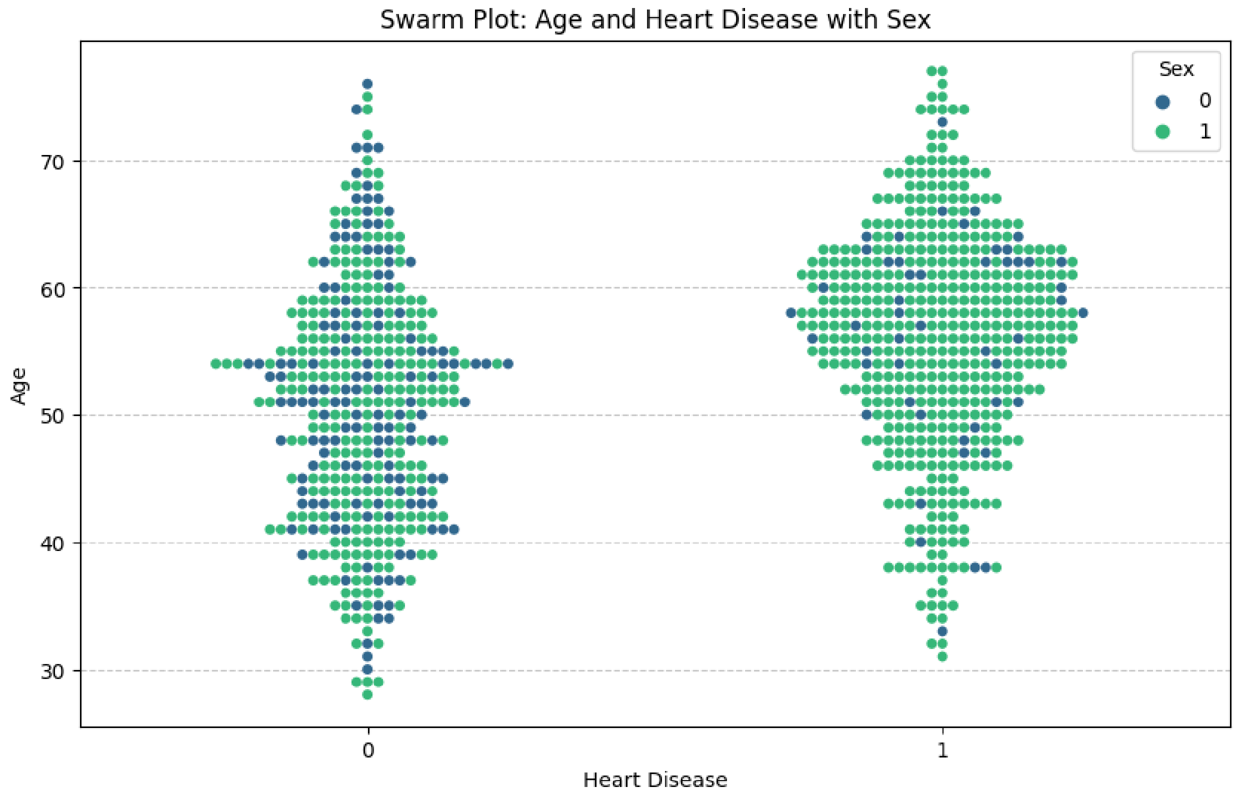


Figure 14: Swarm Plot: Age and Heart Disease with Sex

The diagram above shows the distribution of the Heart Disease with relation to Age and Sex.

- Each point on the plot represents an individual in the dataset.
- The color of the points distinguishes between individuals with and without heart disease.

Notably, the swarm plot reveals a higher concentration of younger individuals in the non-heart disease category, while individuals with heart disease exhibit a broader age distribution.

Col 2: Swarm Plot: Age and Sex with Heart Disease

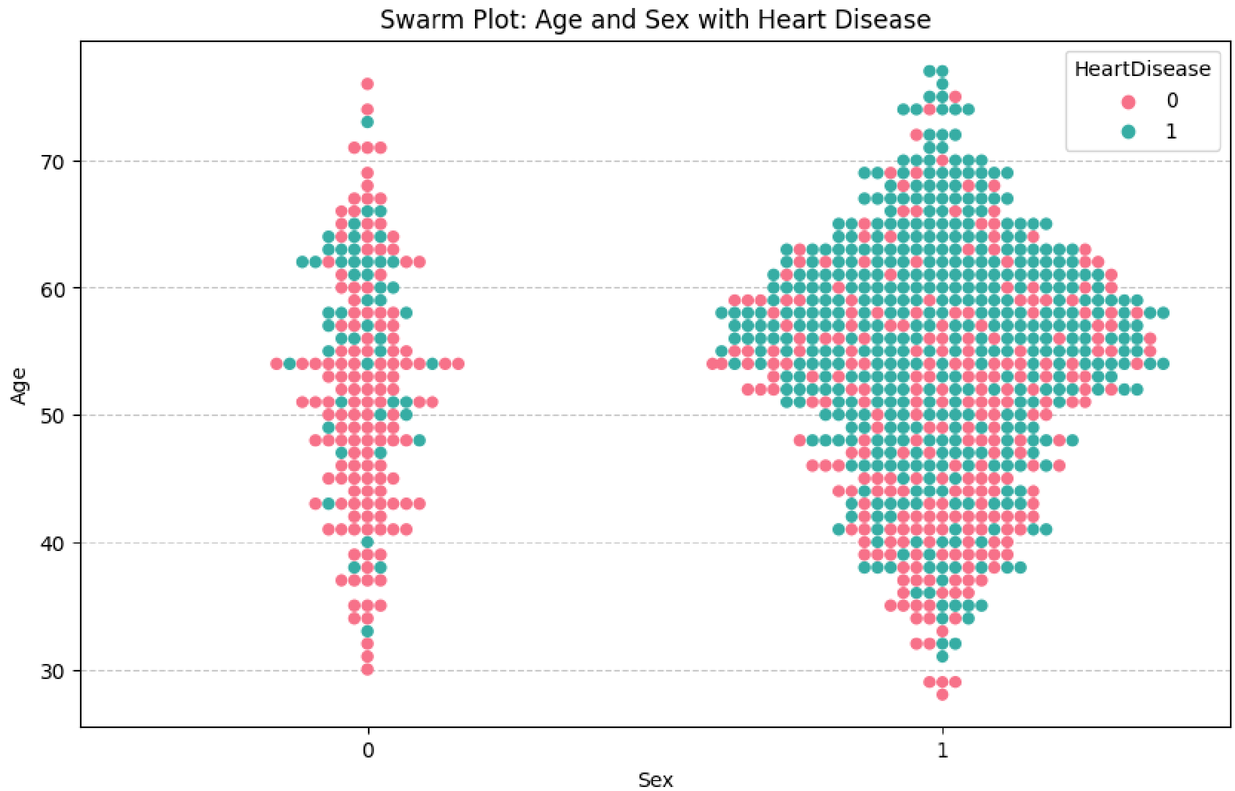


Figure 15: Swarm Plot: Age and Sex with Heart Disease

The distribution of Sex with relation to Age and Heart Disease is depicted in the above diagram. It can be observed that the number of males is far greater than the number of females. Additionally, the ratio of males with heart disease is greater than the ratio of males without heart disease. On the contrary, the ratio of females who are healthy is greater than the ratio of females with heart disease.

Pie Chart: Relationship between (Sex) and (HeartDisease)

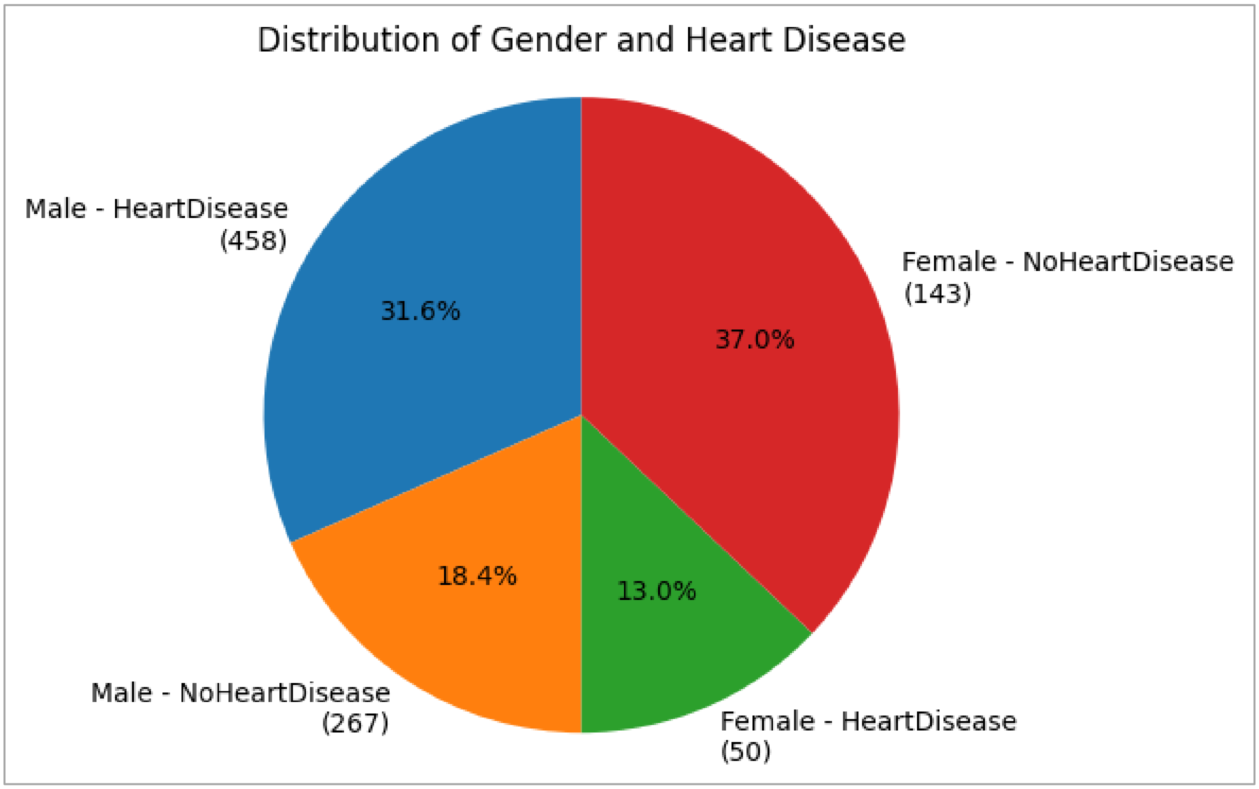


Figure 16: Pie Chart: Relationship between (Sex) and (HeartDisease)

The pie chart shows the distribution of heart disease within male and female populations. Among males, 31.8% have been diagnosed with heart disease, while 22.7% do not have heart disease. In the female group, 27.3% have heart disease, and 18.2% do not.

Count of Males and Females by Heart Disease Status

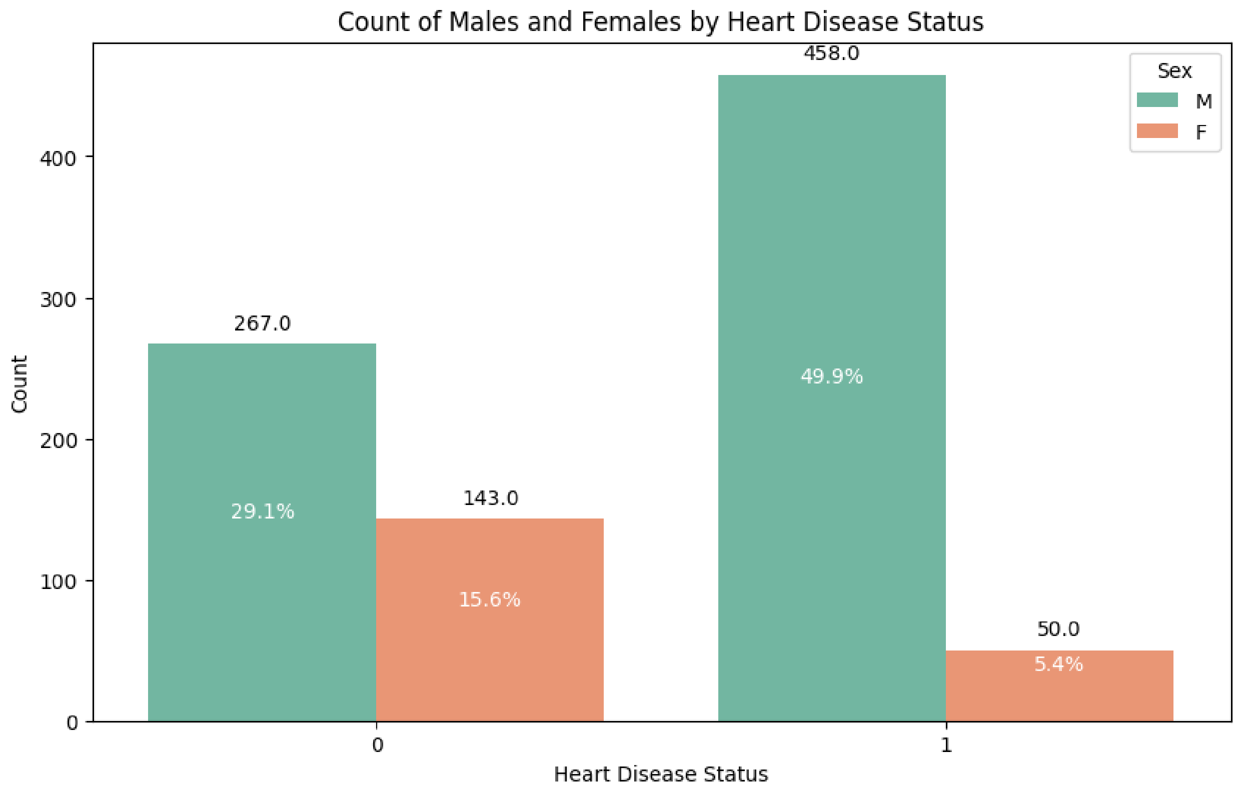


Figure 17: Count of Males and Females by Heart Disease

The histogram above displays the count of males and females with heart disease and those without. It can be observed that 49.9% of overall males have heart disease, and 29.1% do not have it. The count of males with heart disease is 458, and those without it are 267.

Similarly, it can be seen that 5.4% of all overall females have heart disease, and 15.6% of females do not have it. The count of females with heart disease is 50, and those without it are 143.

Column 3: ChestPainType

Visualizing the distribution of Chest Pain Types

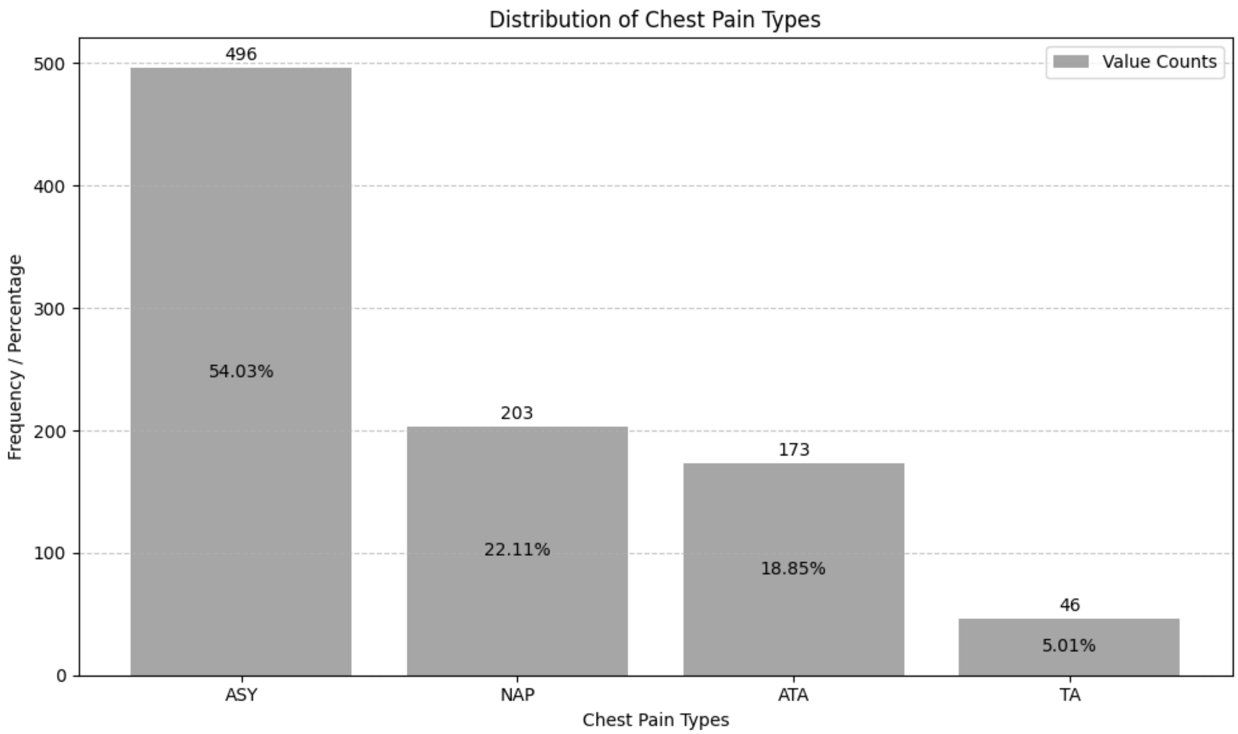


Figure 18: Distribution of counts and percentages of "Chest Pain Types"

The bar plot illustrates the distribution of chest pain types. Among the four categories, 'ASY' (Asymptomatic) stands out as the most prevalent, comprising 54.03% of the observations with a count of 496. Following closely is 'NAP' (Non-Anginal Pain) at 22.11%, represented by a count of 203.

The third category, 'ATA' (Atypical Angina), constitutes 18.85% of the dataset with a count of 173. Lastly, 'TA' (Typical Angina) has the smallest presence, accounting for 5.01% of the data with a count of 46.

This breakdown provides valuable insights into the relative frequencies of different chest pain types, helping to understand their part in the context of heart disease.

Distribution of Chest Pain Types with Heart Disease

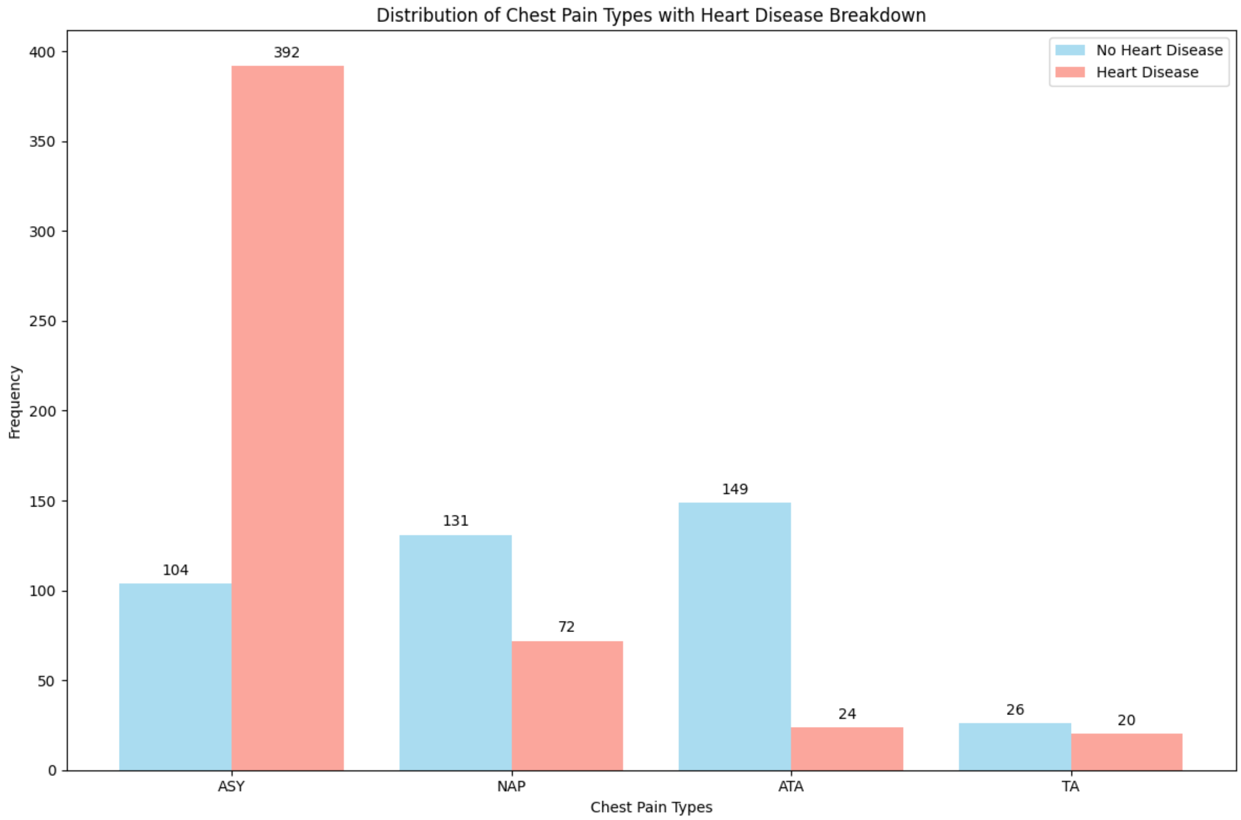


Figure 19: Distribution of Chest Pain Types with Heart Disease

The figure above shows the distribution of chest pain types. ASY has the greatest number of heart disease (392), followed by NAP (72), ATA (24) and TA (20). Whereas, the chest pain type ATA (149) has most amount of people without heart disease, followed by NAP (131), ASY (104) and TA (26).

Swarm Plot of Age and Chest Pain Types with Heart Disease

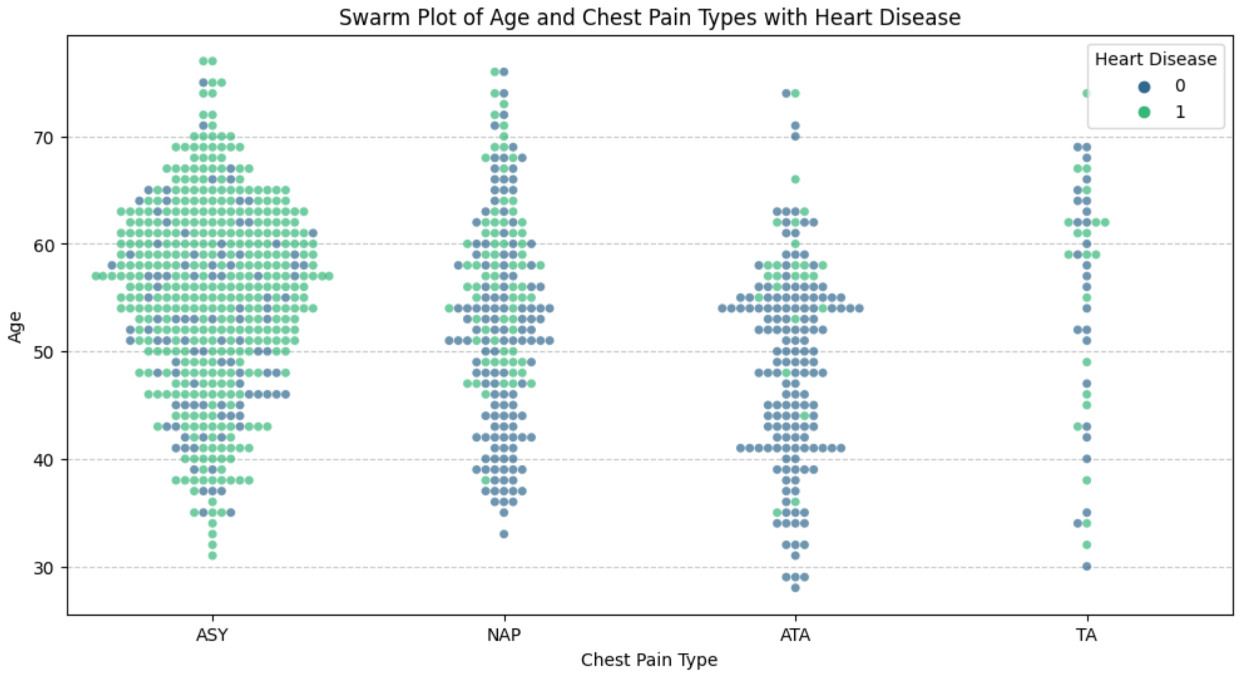


Figure 20: Swarm Plot of Age and Chest Pain Types with Heart Disease

The distribution of Chest pain types with relation to Age and Heart Disease is depicted in the above diagram. The colored data points can be observed, and it can be noticed that 'ASY' is the most common type of chest pain, and it appears to lead to heart disease most of the time. However, in the cases of NAP, ATA, and TA, their occurrences are decreasing linearly, and so is their ability to cause heart disease.

Column 5: Cholesterol (Heart Disease analysis based on Cholesterol)

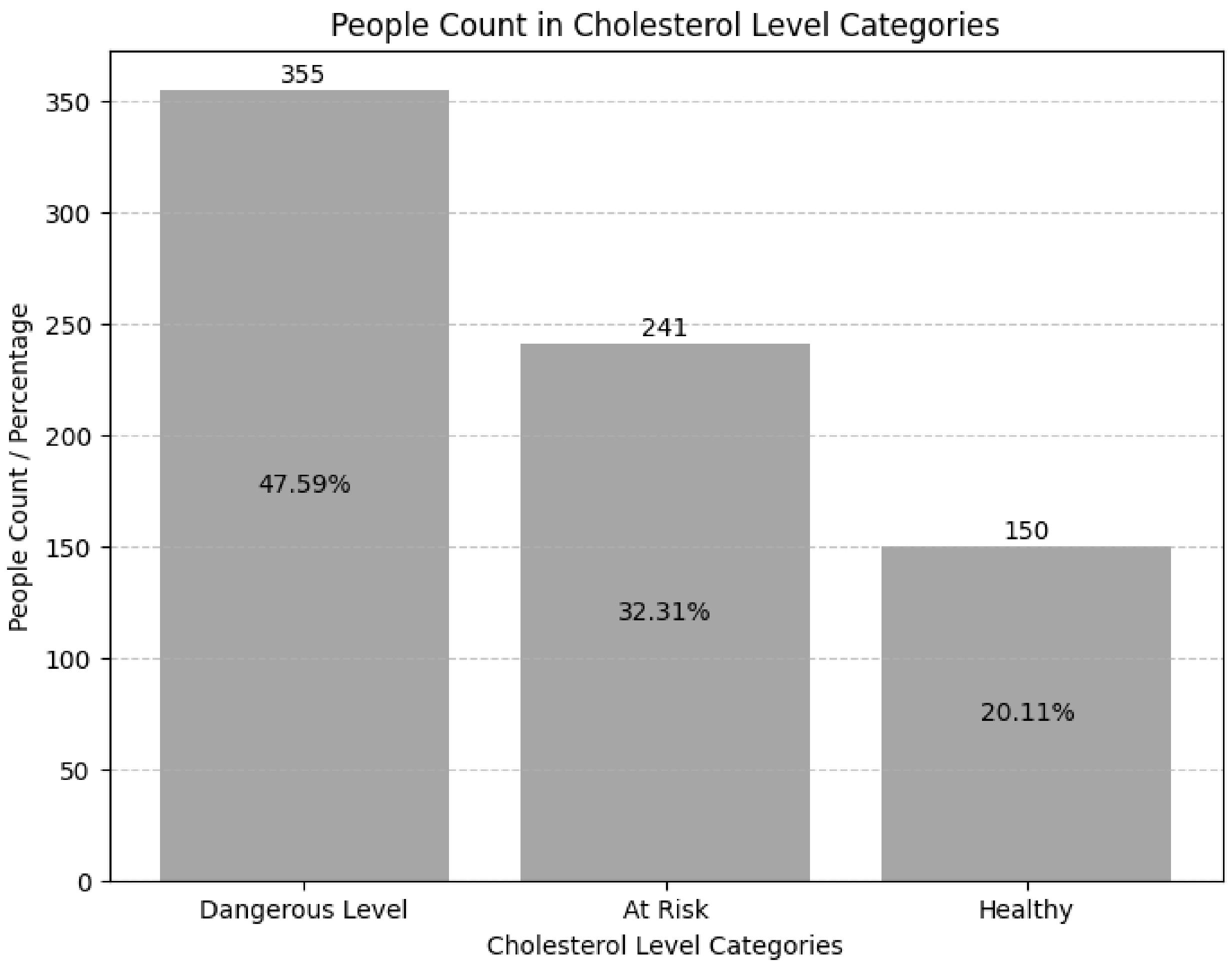


Figure 21: People Count in Cholesterol Level Categories

In medicine, the cholesterol level in adults below 200 mg/dl is considered to be healthy, the level between 200–240 mg/dl is considered to be "At Risk," and the level beyond 240 mg/dl is considered to be at a "Dangerous Level." It can be observed from the figure above that the count of healthy is 150 at 20.11%, the count of "At Risk" is 241 at 32.31%, and the count of "Dangerous Level" is 355 at 47.59%.

These findings offer valuable insights into the distribution of cholesterol levels, and provides a basis for further medical diagnosis.

Column 7: Resting ElectroCardioGram (Heart disease analysis based on ElectroCardioGram)

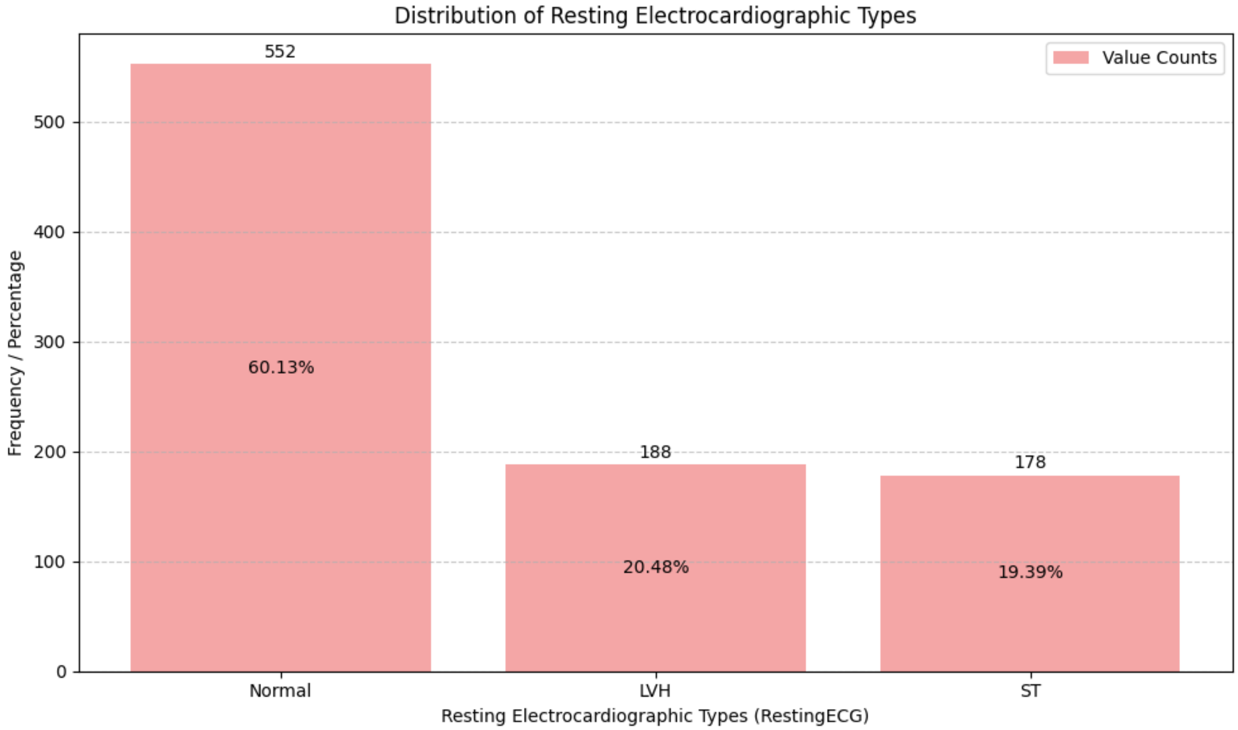
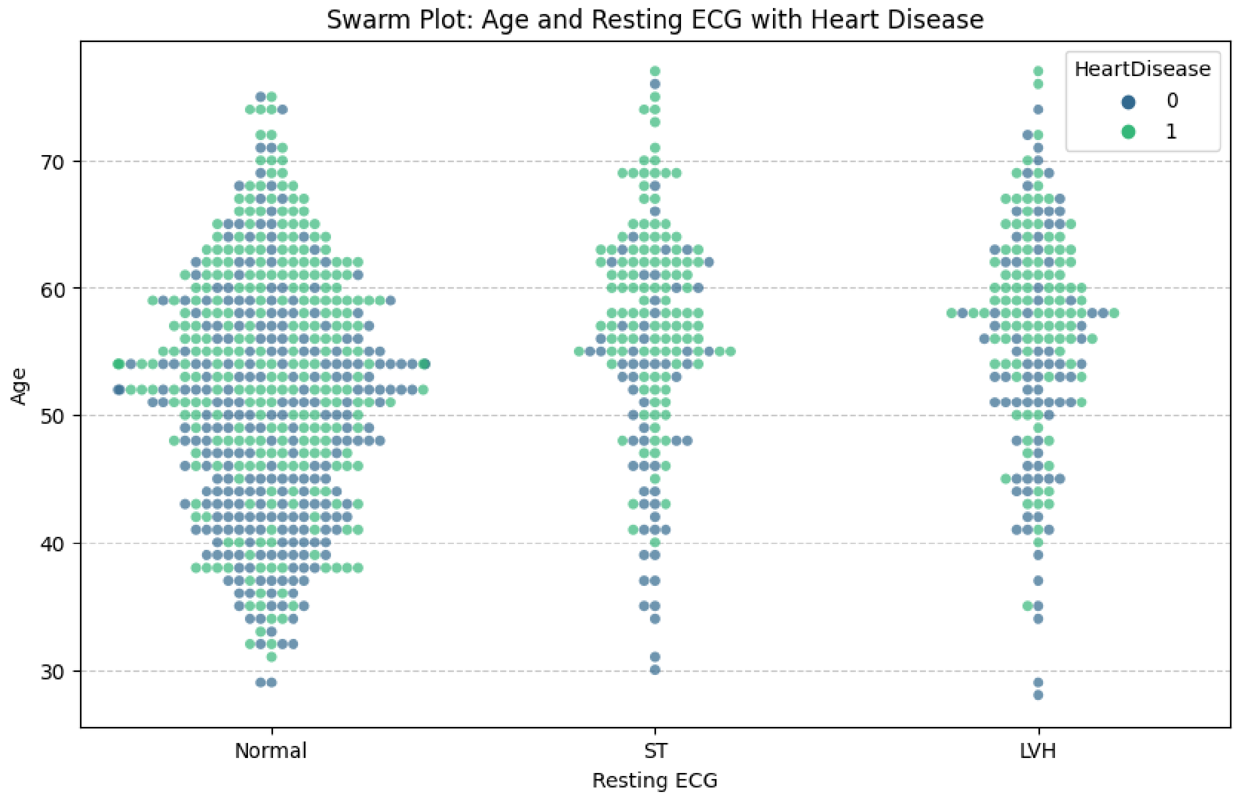


Figure 22: Distribution of Resting Electrocardiographic Types

The histogram above shows the distribution of the types of electrocardiogram (ECG). The most common type of ECG is ‘Normal’, followed by ‘LVH’ and ‘ST’ with their respective value counts as 552 (60.13%), 188 (20.48%) and 178 (19.39%).

These readings help to understand the status of the heart and helps to diagnose abnormal heart rhythms and coronary heart disease (heart attack and angina).

Swarm Plot: Age and Resting ECG with Heart Disease



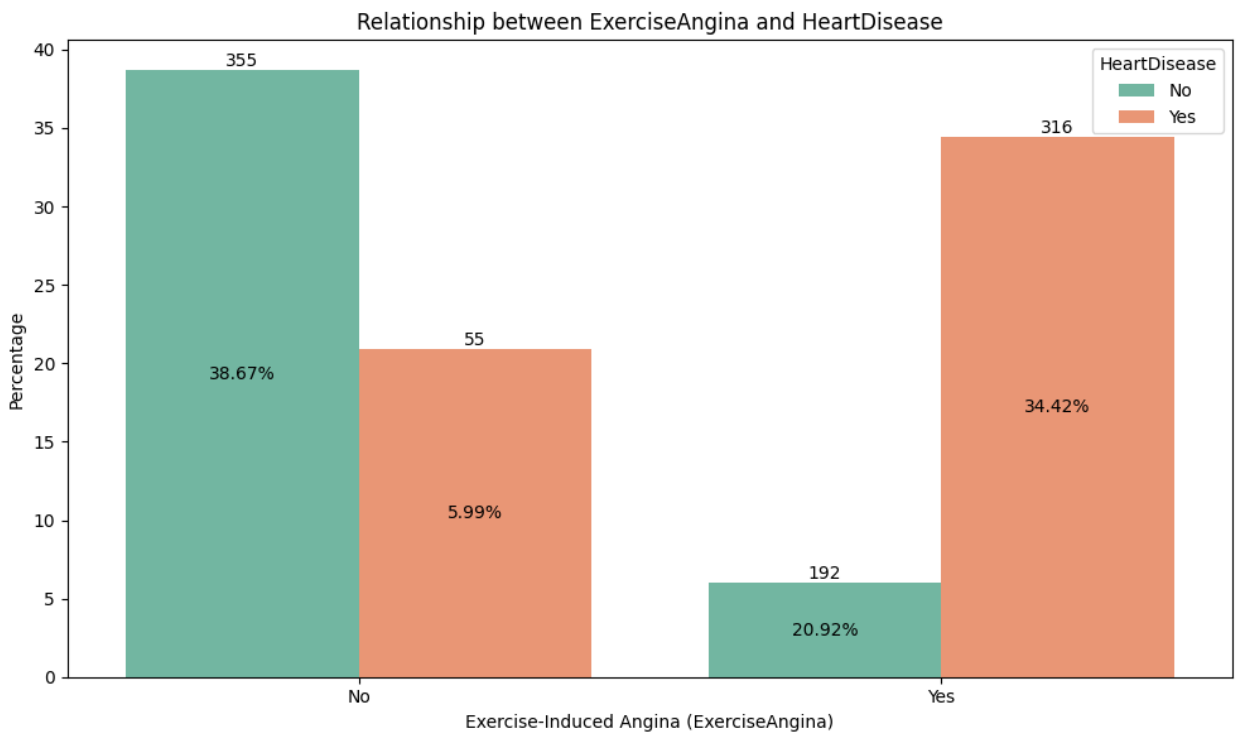
The distribution of Resting ECG and Age with Heart Disease is depicted in the swarm plot above. The most scattered data points are observed on the Normal ECG. It can be seen that older people under Normal ECG have more heart disease. Similarly, it can be observed that older people in LVH and ST also have relatively more heart disease than younger people.

Column 9: Exercise Angina

Exercise-induced angina is a type of *chest pain or discomfort* that occurs when the heart muscle doesn't receive enough blood flow and oxygen during physical exertion or stress. This typically happens during activities like exercise or emotional stress when the heart needs to work harder.

The swarm plot below shows the distribution of Exercise-induced angina with Age and Heart Disease. It is observable that whoever has Exercise-induced angina is highly susceptible to heart disease. One who does not suffer from Exercise-induced angina are also likely to suffer from heart disease but such probability is very low.

Relationship between ExerciseAngina and HeartDisease



The figure above shows the distribution of Exercise Induced Angina (pain) in relation to heart disease. Of those who have Exercise Induced Angina (Yes), 316 (34.42%) have heart disease and 192 (20.92%) do not. Similarly, of those who do not have Exercise Induced Angina (No), 55 counts have heart disease and 355 counts do not have heart disease.

Swarm Plot of Exercise-Induced Angina

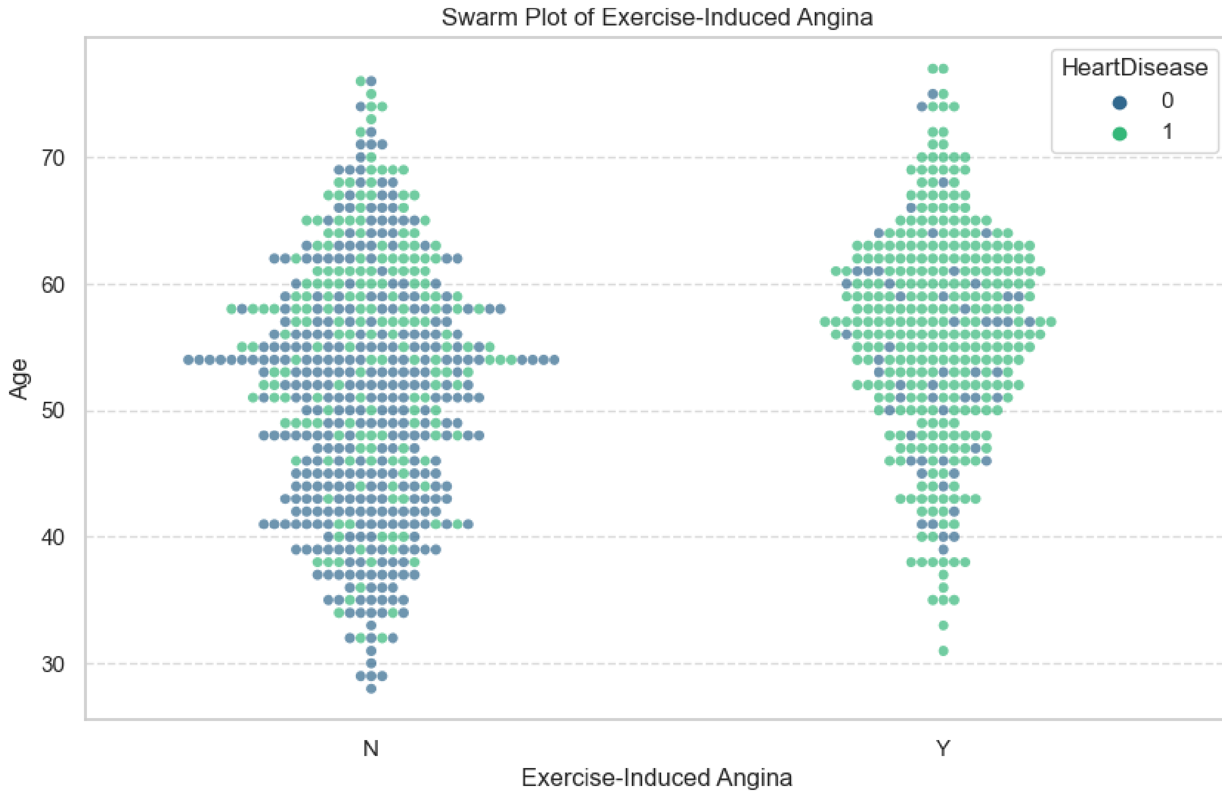


Figure 23: Swarm Plot of Exercise-Induced Angina

Column 11: ST_Slope

The ST_Slope typically refers to a feature or variable that represents the slope of the ST segment on an electrocardiogram (ECG or EKG). The ST segment is a portion of the ECG waveform that reflects the period between ventricular depolarization and repolarization.

The ST segment slope is often used as a diagnostic indicator in assessing cardiac health. Changes in the ST segment can provide valuable information about myocardial ischemia or injury. The three common types of ST segment slopes are:

1) **Upsloping (Positive Slope):**

   This may be considered normal in some cases, but significant upsloping can be indicative of myocardial ischemia.

2) Down sloping (Negative Slope):

Down sloping ST segments can be associated with certain types of heart conditions, including myocardial infarction (heart attack) or ischemia.

3) Flat (Horizontal):

A flat ST segment may also be abnormal and could suggest myocardial ischemia.

Distribution of ST Slope Types

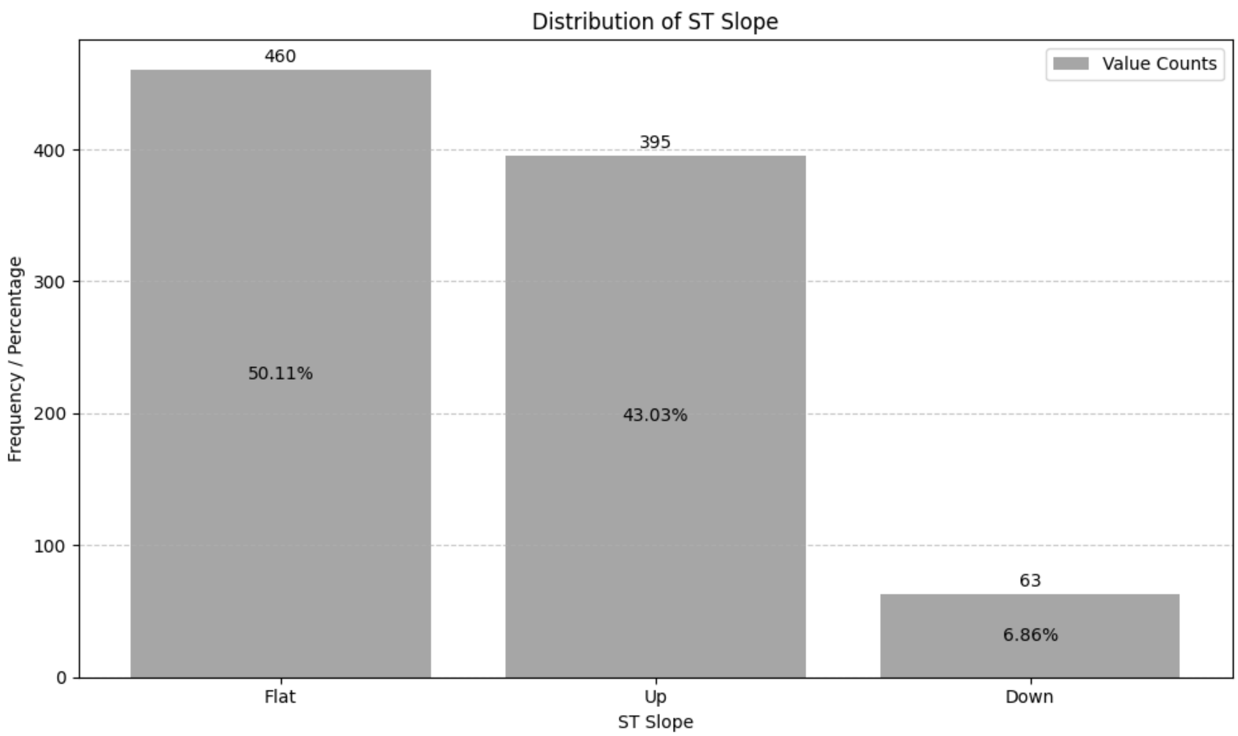


Figure 24: Distribution of ST Slope Types

The histogram above shows the distribution of three common types of ST_Slope. It is noticeable that most people have Flat slope constituting 460 count (50.11%). *To see how many people under Flat slope have heart disease, do see the swarm plot below.*

Similarly, the UP slope has the second highest value count of 395 (43.03%). Lastly, the Down slope has the least value count of 63 (6.86%).

Swarm Plot of Age and ST_Slope with Heart Disease

The swarm plot below shows the number of people who have or do not have heart disease in each of the three slopes (Up, Flat and Down). It is noticeable that Flat slope is the most dangerous segment as the majority of people present in this slope have heart disease.

In the Up slope (as mentioned above) is considered normal but significant upsloping can also lead to heart disease. This has been reflected below in people of age between 55 and above.

The Down slope also indicates abnormal state. Even though, the total number of Down slopes is small, anyone present in this segment, has a very high probability of having heart disease.

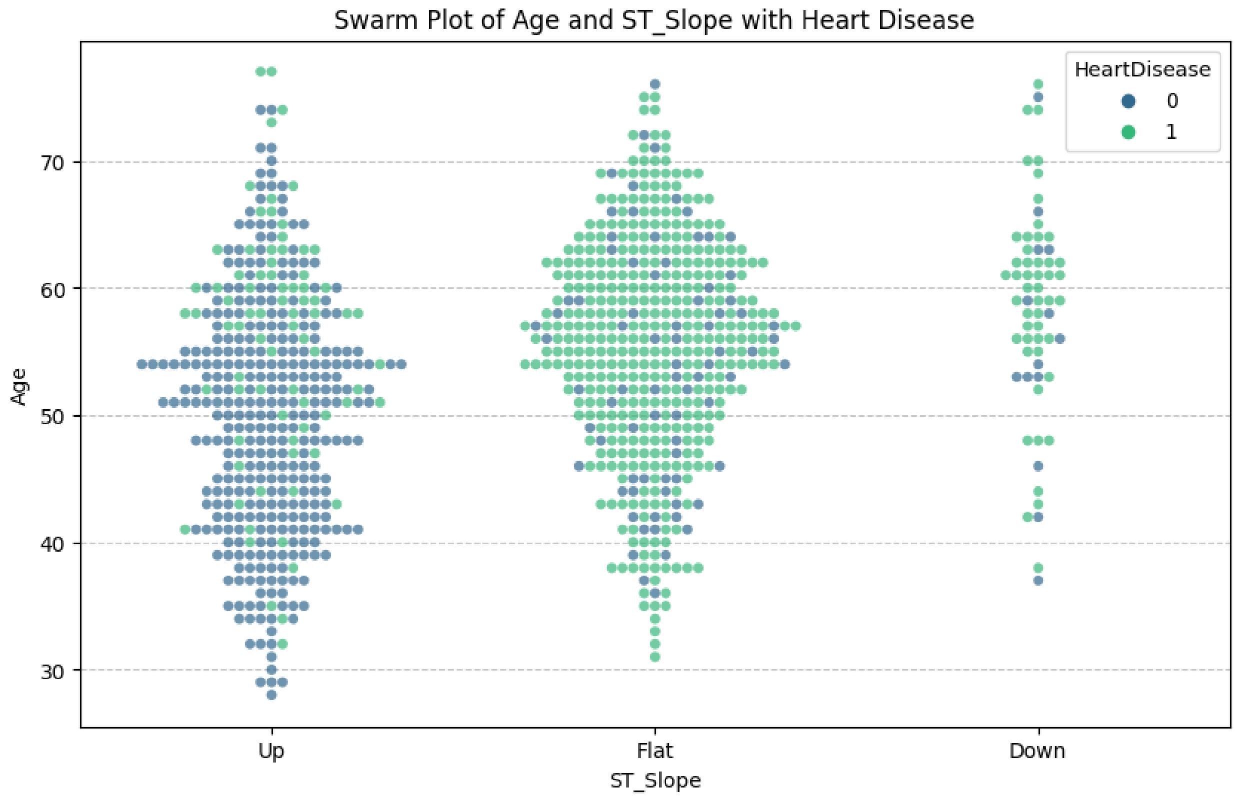


Figure 25: Swarm Plot of Age and ST_Slope with Heart Disease

Violin Plot of Age and ST_Slope

Figure 26: Violin Plot of Age and ST_Slope

KDE (Kernel Density Estimation) Distribution of All Features

Kernel Density Estimation (KDE) is a statistical technique used to visualize the probability density function of a dataset. When applied to features in a dataset, KDE helps assess skewness, which is the lack of symmetry in a distribution.

By plotting the smooth curve of the KDE, it can be determined whether the distribution is skewed to the right (positively skewed) or skewed to the left (negatively skewed). Skewed distributions exhibit a longer tail in one direction, and the direction of skewness can be indicated by the position of the peak (mode) in relation to the median and mean. (Yolyan, 2022)

In a skewed distribution, the mode (peak) of the KDE plot may not align with the median and mean.

Positive skewness: Mode < Median < Mean

Negative skewness: Mean < Median < Mode

KDE provides an effective visual tool for comparing the skewness of different features in a dataset and is useful for exploratory data analysis.

In the diagrams shown below, it can be noticed that the distribution in -

- RestingBP is negatively skewed as it has longer tail towards left direction
- Cholesterol is positively skewed as it has longer tail towards right direction.
- Oldpeak seems to be both positively and negatively skewed
- Other features like Age, Sex, ChestPainType, FastingBS, RestingECG, ExerciseAngina and ST_Slope are relatively symmetric.

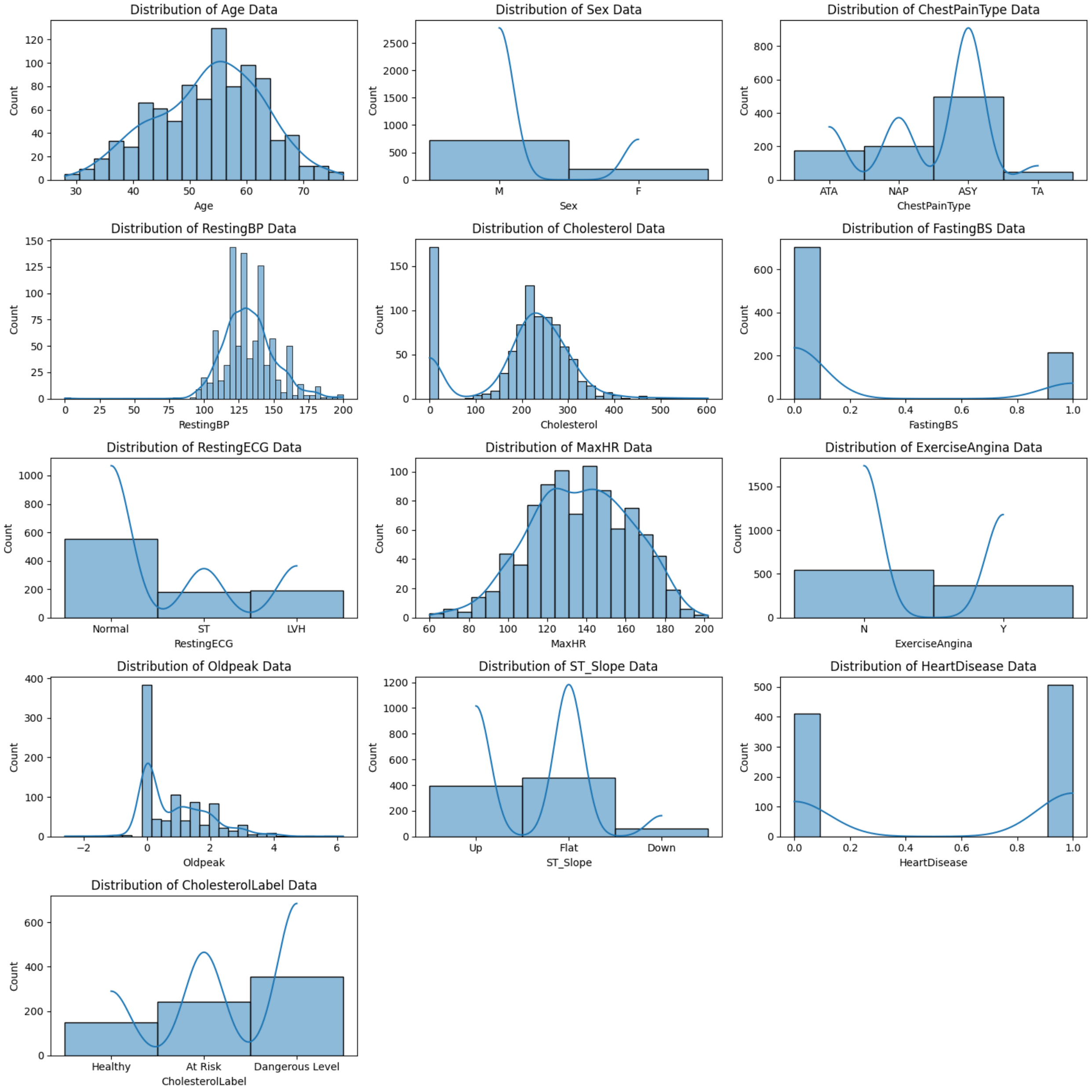


Figure 27: Histogram Distribution of All Features

Heatmap

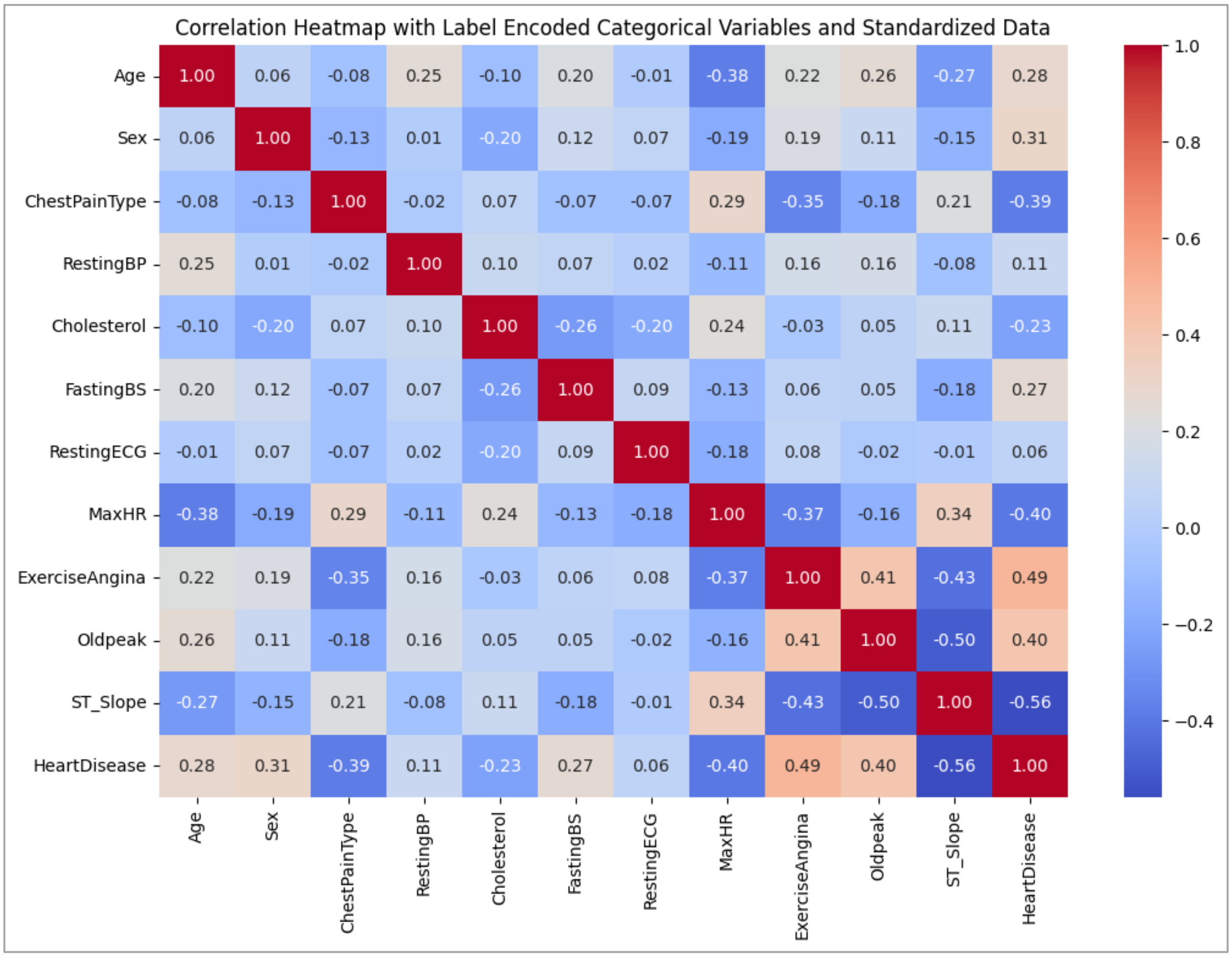


Figure 28: Heatmap

Interpreting a heatmap involves analyzing the correlation coefficients between different variables. In the context of the heart disease dataset, a correlation coefficient close to 1 indicates a strong positive correlation, close to -1 indicates a strong negative correlation, and close to 0 indicates a weak correlation.

Age (0.28): There is a weak positive correlation between age and the likelihood of heart disease. As age increases, there is a slight tendency for the risk of heart disease to increase.

Sex (0.31): There is a weak positive correlation between sex and heart disease. It suggests that being male may be associated with a slightly higher risk of heart disease compared to being female.

ChestPainType (-0.39): There is a moderate negative correlation between chest pain type and heart disease. A higher negative value indicates that certain types of chest pain might be associated with a lower risk of heart disease.

RestingBP (0.11): There is a weak positive correlation between resting blood pressure and heart disease. However, the correlation is not very strong, suggesting that other factors play a more significant role.

Cholesterol (-0.23): Heart disease and cholesterol levels have a marginally negative association.. Lower cholesterol levels are associated with a slightly higher risk.

FastingBS (0.27): There is a weak positive correlation between fasting blood sugar and heart disease. Higher fasting blood sugar levels might be associated with a slightly higher risk.

RestingECG (0.06): There is a very weak positive correlation between resting electrocardiographic results and heart disease.

MaxHR (-0.40): There is a moderate negative correlation between maximum heart rate achieved and heart disease. A lower maximum heart rate might be associated with a higher risk.

ExerciseAngina (0.49): There is a moderate positive correlation between exercise-induced angina and heart disease. Presence of exercise-induced angina is associated with a higher risk.

Oldpeak (0.40): Heart disease and depression brought on by exercise compared to rest have a somewhat positive link.

ST_Slope (-0.56): There is a strong negative correlation between the slope of the peak exercise ST segment and heart disease. Certain slopes might be associated with a lower risk.

5.2 Data Modeling

Data modeling refers to the process of creating a mathematical representation or model based on the underlying patterns and structures within a dataset. This modeling is a crucial step in building predictive or descriptive models that can make informed decisions.

Data modeling in machine learning encompasses various techniques, including regression, classification, clustering, and deep learning. The choice of modeling technique depends on the nature of the data and the specific problem the model aims to address. The process may also involve feature engineering, where relevant features are selected or transformed to improve the model's performance.

Data Pre-processing

Data modeling in machine learning involves various techniques, including regression, classification, clustering, and deep learning. The choice of modeling technique depends on the nature of the data and the specific problem the model aims to address. The process may also involve feature engineering, where relevant features are selected or transformed to improve the model's performance.

The common techniques of data pre-processing are:

1) Handle missing data
2) Dealing with outliers
3) Feature Scaling
4) Encoding categorical variables
5) Feature engineering
6) Data splitting

The removal of missing values (NaN or null) in each column of the data frame can be accomplished using the isnull() function. The sum of missing values from every column is calculated, and its count is provided.

Similarly, the duplicate values can also be checked and removed.

Missing Values Detection

In [123]:
```
data.isnull().sum()
```

Out[123]:
```
Age                   0
Sex                   0
ChestPainType         0
RestingBP             0
Cholesterol           0
FastingBS             0
RestingECG            0
MaxHR                 0
ExerciseAngina        0
Oldpeak               0
ST_Slope              0
HeartDisease          0
CholesterolLabel    172
dtype: int64
```

In [124]:
```
data.duplicated().sum()
```

Out[124]:
```
0
```

Transforming Categorical variables into Integer using LabelEncoder

Label encoding is employed to convert categorical (text-based) data into a numerical format in a DataFrame. For each column containing text data, a for loop is utilized to check, and for each unique category in that column, a unique number is assigned to represent it. This process is considered helpful because many machine learning algorithms operate with numerical data, and the conversion of categorical data into numbers enables the effective use of these algorithms. The process, known as Label Encoding, is available in the scikit-learn library.

```
for i in data.select_dtypes(include=['object']).columns:
    labelencoder = LabelEncoder()
    data[i] = labelencoder.fit_transform(data[i])
data.head()
```

	Age	Sex	ChestPainType	RestingBP	Cholesterol	FastingBS	RestingECG	MaxHR	ExerciseAngina	Oldpeak	ST_Slope	HeartDisease
0	40	1	1	140	289	0	1	172	0	0.0	2	0
1	49	0	2	160	180	0	1	156	0	1.0	1	1
2	37	1	1	130	283	0	2	98	0	0.0	2	0
3	48	0	0	138	214	0	1	108	1	1.5	1	1
4	54	1	2	150	195	0	1	122	0	0.0	2	0

Checking if the Data types has been Transformed

Using the dtypes() function, it can be noticed that all of the data types have been transformed into integer.

```
data.dtypes
```

```
Age                  int64
Sex                  int32
ChestPainType        int32
RestingBP            int64
Cholesterol          int64
FastingBS            int64
RestingECG           int32
MaxHR                int64
ExerciseAngina       int32
Oldpeak            float64
ST_Slope             int32
HeartDisease         int64
CholesterolLabel  category
dtype: object
```

Separating the Independent and Target variables as X and y

The features (independent variables) from Age column to ST_Slope have been separated and stored in X. Similarly, y is used to store the target variable that the model aims to predict.

```
# Separating the X and y columns before splitting the dataset. y is the target column
X, y = data.loc[:, :'ST_Slope'], data['HeartDisease']
```

```
X.head()
```

	Age	Sex	ChestPainType	RestingBP	Cholesterol	FastingBS	RestingECG	MaxHR	ExerciseAngina	Oldpeak	ST_Slope
0	40	1	1	140	289	0	1	172	0	0.0	2
1	49	0	2	160	180	0	1	156	0	1.0	1
2	37	1	1	130	283	0	2	98	0	0.0	2
3	48	0	0	138	214	0	1	108	1	1.5	1
4	54	1	2	150	195	0	1	122	0	0.0	2

Splitting our dataset into Training and Testing sets

The dataset is split into training and testing sets using the train_test_split() function from scikit-learn. The sets that are left over are y_train, y_test, X_train, and X_test.

X and **y**: The features and target variable.

random_state: Ensures that the split will be the same each time the code is run.

test_size: This case, 30% of the data will be used for testing.

shuffle: If True, the data will be randomly shuffled before splitting.

```
# Splitting our dataset into training and testing sets in 70:30 ratio

X_train, X_test, y_train, y_test = train_test_split(X, y, random_state=40, test_size=0.3, shuffle=True)

# Display the shapes of the resulting sets
print("X_train shape:", X_train.shape)
print("X_test shape:", X_test.shape)
print("y_train shape:", y_train.shape)
print("y_test shape:", y_test.shape)
```

```
X_train shape: (642, 11)
X_test shape: (276, 11)
y_train shape: (642,)
y_test shape: (276,)
```

Data Standardization

Data standardization, also known as feature scaling, is a preprocessing step in machine learning that involves transforming the numerical features of a dataset to a standard scale. The goal is to ensure that different features are on a similar scale and to prevent some features from dominating others due to differences in their magnitudes.

This process is particularly important for algorithms that are sensitive to the scale of the input features, such as support vector machines, k-nearest neighbors, and neural networks.

Standardizing the data ensures that the model is not influenced by the inherent scale of the features, allowing it to learn meaningful patterns without being biased by the magnitude of the input values.

Z-score normalization (Standardization)

The Z-score method has been employed, which scales the features to have a mean of 0 and a standard deviation of 1. It is calculated as (x - mean) / standard deviation, where x represents the original value.

```python
# Standardize the data
scaler = StandardScaler()
X_train_standardized = scaler.fit_transform(X_train)
X_test_standardized = scaler.transform(X_test)
X_train_standardized
```

```
array([[-0.35981154,  0.50145986, -0.8090179 , ...,  1.20431793,
         3.86701951, -0.62600258],
       [ 0.99288536,  0.50145986, -0.8090179 , ...,  1.20431793,
        -0.7937895 , -0.62600258],
       [-0.15170433,  0.50145986, -0.8090179 , ...,  1.20431793,
         0.1383723 , -0.62600258],
       ...,
       [ 0.05640289,  0.50145986,  0.24451048, ..., -0.83034552,
        -0.7937895 ,  1.03471501],
       [-2.5449373 ,  0.50145986,  0.24451048, ..., -0.83034552,
        -0.7937895 ,  1.03471501],
       [-0.88007958,  0.50145986,  1.29803886, ..., -0.83034552,
        -0.88700568,  1.03471501]])
```

5.3 Models

Decision Trees, KNN, and Random Forest have been utilized, and their accuracy, precision, recall, F1-score, and the confusion matrix have been calculated. To enhance the performance of the models, K-Fold cross-validation and Search Grid CV have been employed. The scores achieved after applying these optimization techniques are presented below in three stages.

5.3.1 K Fold Cross Validation

K-Fold Cross Validation is a popular technique in machine learning for assessing the performance and generalizability of a model. K-Fold Cross Validation is like a smart way to check how good a machine learning model is.

Imagine you have a pile of cards, and you want to see how well you can play a game with them. Instead of using the entire pile at once, you split it into smaller groups (let's say 5 groups). Now, you play the game five times, each time using a different group of cards to test and the rest for practice. (Brownlee , A Gentle Introduction to k-fold Cross-Validation, 2023)

After the game is played five times, an assessment is made of how well was done each time. This aids in obtaining a better idea of overall proficiency in the game, not just in one specific round. In machine learning, a similar approach is taken. The data is split, with some parts used to train the model and other parts to assess its performance. By repeating this process several times and averaging the results, a more reliable measure of the model's performance is obtained. It's akin to ensuring that the model performs well in the game regardless of the set of cards it receives.

The primary idea behind K-Fold Cross Validation is to partition the original dataset into K equally sized folds or subsets. The model is then trained and evaluated K times, each time using a different fold as the testing set and the remaining data as the training set. (Jha & Lyashenko, 2023)

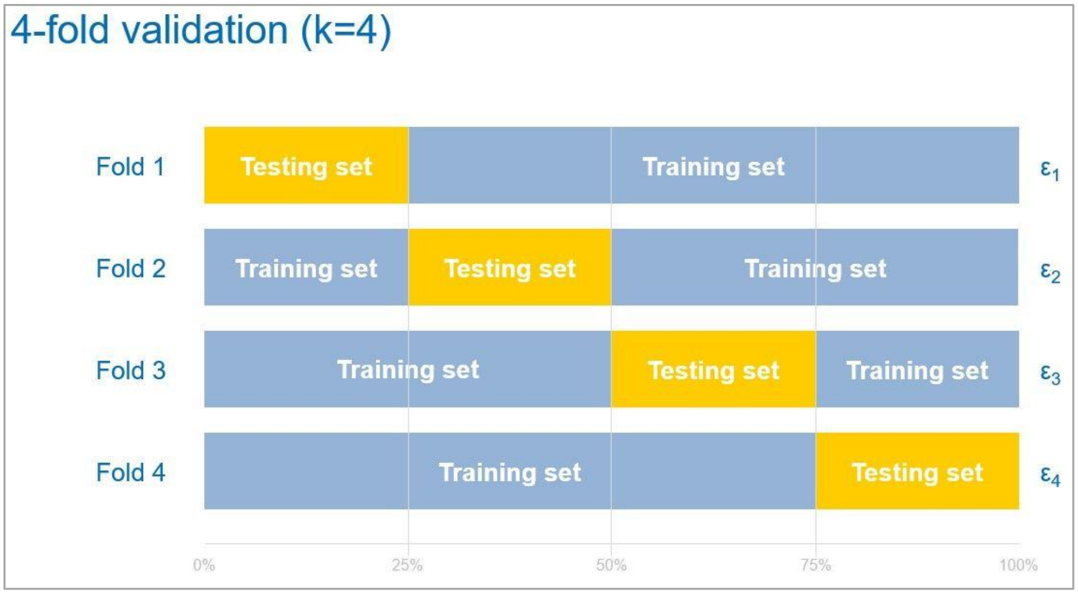


Figure 29: 4-fold cross validation (MathWorks, 2024)

Here's how it typically works:

Partitioning: K folds, or subsets, make up the dataset.

Training and **Testing**: Using a different fold as the testing set and the remaining folds as the training set, the model is trained K times.Performance

Evaluation: The performance of the model is evaluated by calculating the average performance metric across all K iterations. Evaluation metrics include accuracy, precision, recall, and F1-score.

K-Fold Cross Validation helps in obtaining a more robust estimate of a model's performance compared to a single train-test split. It ensures that the model is tested on different subsets of data, providing a more comprehensive assessment of its ability to generalize to new, unseen data. This technique is especially useful when the dataset is limited, and it helps to mitigate issues related to the randomness of a single train-test split.

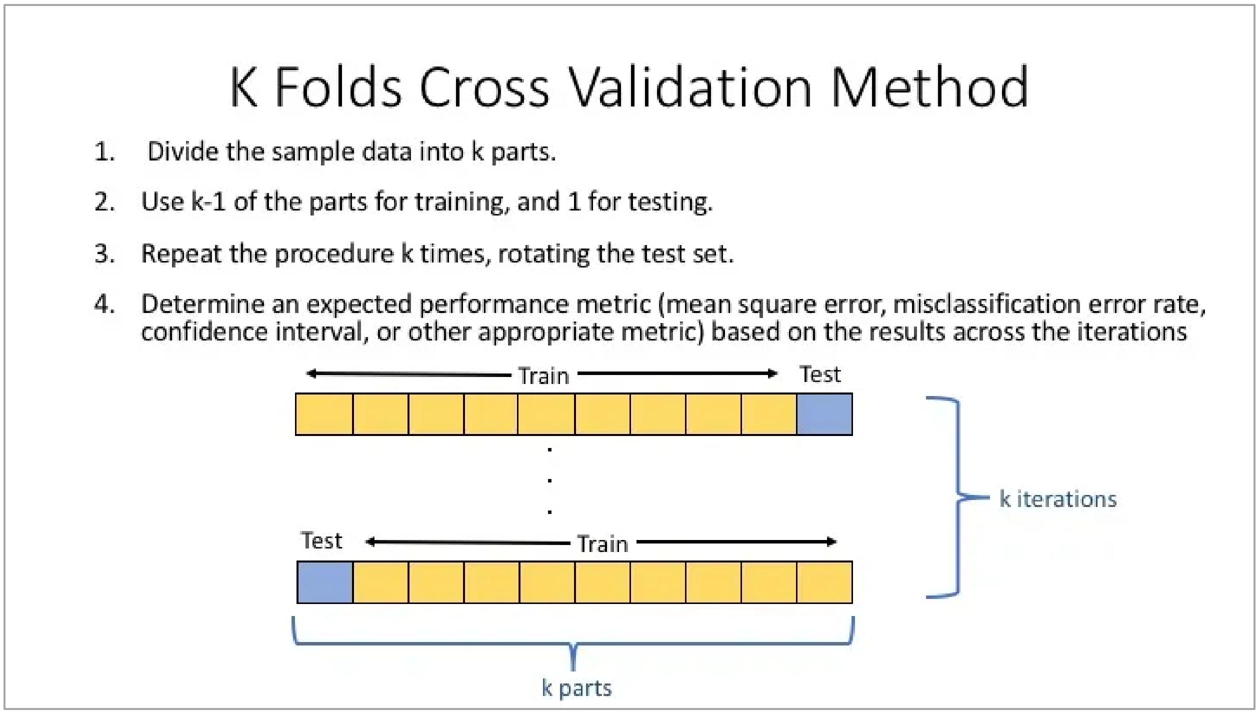


Figure 30: K-Fold Cross Validation Method (Terribile, 2017)

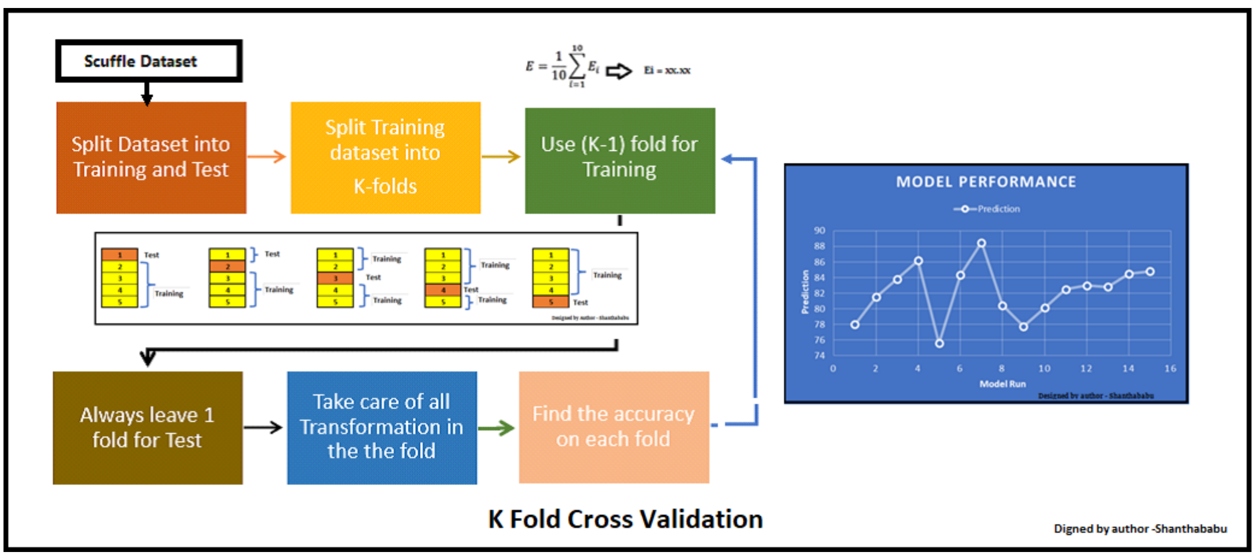


Figure 31: K-Fold Cross Validation (Pandian, 202)

5.3.2 Hyperparameter Tuning

Hyperparameter tuning, often referred to as hyperparameter optimization or hyperparameter search, is the process of finding the best configuration for the hyperparameters of a machine learning model. Hyperparameters are external configuration settings that are not learned from the data but are set before the training process. A neural network's number of hidden layers, regularization strengths, and learning rates are a few examples.

The goal of hyperparameter tuning is to improve the performance of a model by identifying the optimal values for these hyperparameters. This process often involves systematically searching through a predefined space of hyperparameter values, evaluating the model's performance using metrics like accuracy or loss, and selecting the combination of hyperparameters that leads to the best results.

Common techniques for hyperparameter tuning include grid search, random search, and more advanced methods like Bayesian optimization or genetic algorithms. The ultimate aim is to find hyperparameter values that enable the model to generalize well to new, unseen data and achieve optimal performance. (Amazon, 2024)

5.2.3 Search Grid CV

A machine learning technique called Search Grid Cross Validation (CV) is used to systematically search through a predetermined grid of hyperparameter values. Hyperparameters are machine learning algorithm configuration settings that are set before the training process and are not learned from the data.

Search Grid CV is the process of systematically testing different combinations of these parameters to see which one gives the best performance. Instead of randomly guessing, it helps you explore the possibilities in an organized way. It's like having a taste test for various versions of your recipe to find the one that makes your dish (or model) the most flavorful (or accurate). In essence, Search Grid CV is a smart way to optimize your machine learning model for the best results. (Shah, 2023)

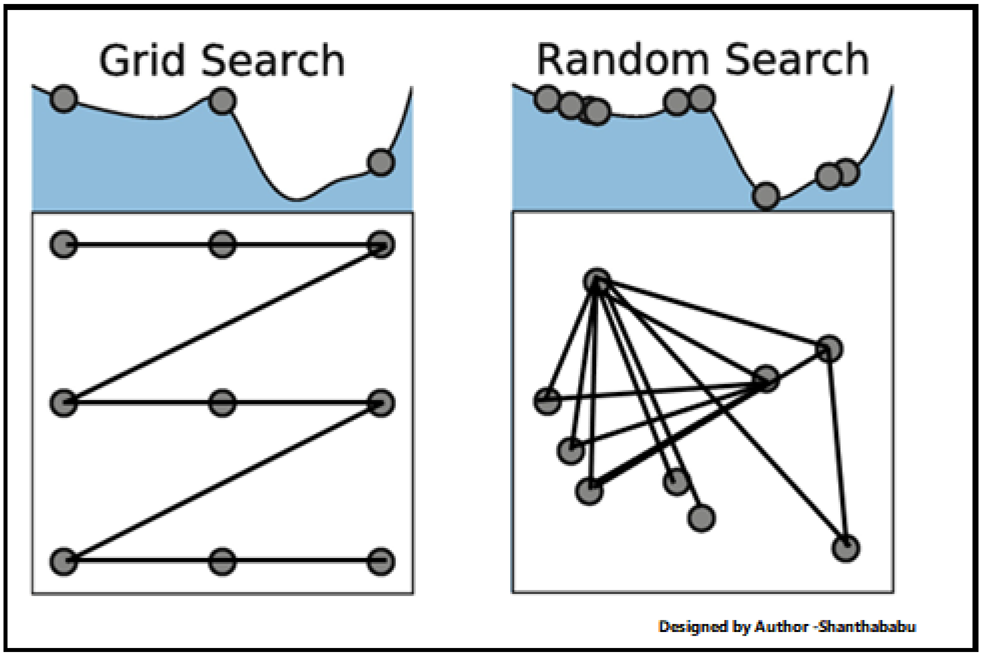


Figure 32: Search Grid CV (Shah, 2023)

In Search Grid CV, a grid of hyperparameter combinations is specified, and the model is trained and evaluated for each combination using cross-validation. Cross-validation involves splitting the dataset into multiple subsets, training the model on a subset, and evaluating its performance on the remaining data. This process is repeated for each combination of hyperparameters in the grid.

The goal is to identify the set of hyperparameters that results in the best model performance, as measured by a specified metric (e.g., accuracy, precision, recall). The search is typically guided by techniques like grid search or random search, exploring the hyperparameter space systematically or randomly, respectively. (Hyperparameter tuning using GridSearchCV, 2024)

Search Grid CV helps in optimizing the model's generalization performance by finding the hyperparameter configuration that performs well across different subsets of the data, reducing the risk of overfitting to a specific dataset or set of parameters.

5.4 Evaluation Metrics

5.4.1 Accuracy

Accuracy, in the context of machine learning, is a performance metric that measures the proportion of correctly classified instances out of the total instances in a dataset. It is calculated by dividing the number of correct predictions by the total number of predictions and is often expressed as a percentage.

Accuracy is like a report card for your model. It tells you how many questions your model got right out of all the questions it tried to answer. If your model predicted 8 out of 10 questions correctly, its accuracy would be 80%. So, the higher the accuracy percentage, the better your model is at giving the right answers. However, remember that it doesn't tell you about the types of mistakes it makes, just the overall number of correct answers.

Accuracy = ((Number of Correct Predictions) / (Total Number of Predictions)) x 100

Here's a breakdown of the terms:

Number of Correct Predictions: The count of instances where the model's prediction matches the actual class labels.

Total Number of Predictions: The sum of correct predictions and incorrect predictions.

The result is usually expressed as a percentage. For example, an accuracy of 85% means that the model correctly predicted the class of 85% of the instances in the dataset.

While accuracy provides a straightforward measure of overall correctness, it may not be sufficient in scenarios where the dataset has imbalanced class distribution. In such cases, other metrics like precision, recall, F1-score, or area under the ROC curve (AUC-ROC) might offer a more nuanced evaluation of the model's performance.

5.4.2 Precision

A machine learning performance metric called precision assesses how effectively a model forecasts successful results. To calculate it, divide the total number of true positive and false positive forecasts by the number of true positive predictions.

Imagine you have a robot friend designed to predict if it's going to be a sunny day. Precision is about how accurate your robot is when it confidently says, "It will be sunny." Precision measures the number of correct positive predictions (it will be sunny) divided by the total number of positive predictions it made (both correct and incorrect).

Precision = (True Positives) / (True Positives + False Positives)

True Positives (TP) are instances correctly predicted as positive.

False Positives (FP) are instances incorrectly predicted as positive.

Precision is particularly relevant in scenarios where the cost of false positive predictions is high. It provides insights into the model's ability to make accurate positive predictions, emphasizing the reliability of positive classifications. Higher precision values indicate a lower rate of false positives, reflecting better performance in correctly identifying positive instances.

5.4.3 Recall

A performance statistic called recall, sometimes referred to as sensitivity or true positive rate, assesses a model's capacity to locate all pertinent instances in a dataset. The ratio of true positive predictions to the total of true positives and false negatives is used to compute it.

Recall is like a measure of a model's memory. It tells us how good the model is at finding all the relevant things in a dataset. If you imagine a model trying to identify dogs in pictures, recall would be a score that shows how many actual dogs it managed to find compared to all the dogs in the pictures. So, high recall means the model is good at not missing out on spotting dogs, and that's important in tasks where you don't want to overlook things.

Recall = True Positives / (True Positives + False Negatives)

True Positives (TP) are instances correctly predicted as positive.

False Negatives (FN) are instances incorrectly predicted as negative.

Recall is particularly important in situations where the cost of false negatives is high, such as in medical diagnoses or critical decision-making scenarios. A higher recall value indicates a lower rate of false negatives, suggesting better performance in capturing all relevant positive instances.

5.4.4 F1 Score

The F1 score is a metric in machine learning that combines both precision and recall into a single value. It is particularly useful when there is an uneven class distribution (e.g., when one class is much more prevalent than the other). The F1 score seeks a balance between precision and recall, providing a comprehensive measure of a model's performance.

F1 Score = 2 x ((Precision x Recall) / (Precision + Recall))

In simpler terms, it's like a compromise between precision and recall – it considers both false positives and false negatives, making it a robust measure of a model's overall effectiveness.

5.5 Stages of evaluating model's performance

Models' performance has been calculated in 3 stages.

5.5.1 Stage 1:

After standardizing the data, calculate the evaluation metrics like accuracy, classification report (precision, recall, F1-score) and confusion metrics. This is before implementing the K-Fold Cross Validation.

5.5.2 Stage 2:

Implement the K-Fold Cross Validation and recalculated the above evaluation metrics.

5.5.3 Stage 3:

Lastly, implement Search Grid Cross Validation and recalculated the above evaluation metrics.

5.6 Models Evaluated

5.6.1 Decision Tree Classifier

Stage 1: Decision Tree Evaluation Metrics before implementing K-Fold Cross Validation

```
Accuracy: 0.77

Classification Report:
              precision    recall  f1-score   support

           0       0.69      0.83      0.75       116
           1       0.85      0.73      0.79       160

    accuracy                           0.77       276
   macro avg       0.77      0.78      0.77       276
weighted avg       0.79      0.77      0.77       276


Confusion Matrix:
 [[ 96  20]
 [ 43 117]]
```

Figure 33: Evaluation Metrics before implementing K-Fold Cross Validation

Accuracy:

The overall accuracy of the Decision Tree model is reported as 0.77 or 77%. This represents the proportion of correctly classified instances out of the total instances in the dataset.

Classification Report:

Precision: Precision is a measure of the accuracy of the positive predictions. For class 0, precision is 0.69, indicating that among the instances predicted as class 0, 69% were correct. For class 1, precision is higher at 0.85, meaning that 85% of instances predicted as class 1 were correct.

Recall (Sensitivity): Recall measures the ability of the model to capture all the relevant instances of a class. For class 0, recall is 0.83, indicating that 83% of actual class 0 instances were correctly predicted. For class 1, recall is 0.73, suggesting that 73% of actual class 1 instances were captured by the model.

F1-score: The F1-score is the harmonic mean of precision and recall. It provides a balance between precision and recall. For class 0, the F1-score is 0.75, and for class 1, it is 0.79.

Support: Support represents the number of actual occurrences of each class in the specified dataset. For class 0, there are 116 instances, and for class 1, there are 160 instances.

Confusion Matrix:

The top-left value (96) represents true negatives (TN), instances correctly predicted as class 0.

The top-right value (20) represents false positives (FP), instances incorrectly predicted as class 1.

The bottom-left value (43) represents false negatives (FN), instances incorrectly predicted as class 0.

The bottom-right value (117) represents true positives (TP), instances correctly predicted as class 1.

Decision Tree Evaluation Metrics after implementing K-Fold Cross Validation

```
Accuracy scores for each fold: [0.80620155 0.79069767 0.765625   0.78125    0.8203125 ]
Average Accuracy across folds: 0.79
Classification Report:
              precision    recall  f1-score   support

           0       0.77      0.78      0.77       294
           1       0.81      0.80      0.81       348

    accuracy                           0.79       642
   macro avg       0.79      0.79      0.79       642
weighted avg       0.79      0.79      0.79       642

Confusion Matrix:
[[229  65]
 [ 68 280]]
```

Figure 34: Evaluation Metrics after implementing K-Fold Cross Validation

Cross-Validation Accuracy:

Accuracy scores for each fold: [0.80620155, 0.79069767, 0.765625, 0.78125, 0.8203125]

Average Accuracy across folds: 0.79 or 79%

In cross-validation, the model is trained and evaluated multiple times on different subsets of the data. The average accuracy across these folds provides a more robust estimate of the model's performance.

Classification Report:

Precision: For class 0, precision is 0.77, indicating that 77% of instances predicted as class 0 were correct. For class 1, precision is 0.81, indicating that 81% of instances predicted as class 1 were correct.

Recall (Sensitivity): For class 0, recall is 0.78, suggesting that 78% of actual class 0 instances were correctly predicted. For class 1, recall is 0.80, indicating that 80% of actual class 1 instances were captured by the model.

F1-score: The F1-score, the harmonic mean of precision and recall, is provided for each class. For class 0, the F1-score is 0.77, and for class 1, it is 0.81.

Support: Support represents the number of actual occurrences of each class in the specified dataset. For class 0, there are 294 instances, and for class 1, there are 348 instances.

Confusion Matrix:

True Negatives (TN): 229 instances were correctly predicted as class 0.

False Positives (FP): 65 instances were incorrectly predicted as class 1.

False Negatives (FN): 68 instances were incorrectly predicted as class 0.

True Positives (TP): 280 instances were correctly predicted as class 1.

Decision Tree Evaluation Metrics after implementing Search Grid Cross Validation

```
Best Parameters: {'criterion': 'gini', 'max_depth': None, 'min_samples_leaf': 1, 'min_samples_split': 10}

Classification Report (Grid Search):
              precision    recall  f1-score   support

           0       0.94      0.96      0.95       294
           1       0.96      0.95      0.95       348

    accuracy                           0.95       642
   macro avg       0.95      0.95      0.95       642
weighted avg       0.95      0.95      0.95       642

Confusion Matrix (Grid Search):
[[281  13]
 [ 19 329]]
```

Figure 35: Decision Tree Evaluation Metrics after implementing Search Grid Cross Validation

Best Hyperparameters:

Best Parameters: {'criterion': 'gini', 'max_depth': None, 'min_samples_leaf': 1, 'min_samples_split': 10}

criterion: For measuring the quality of a split. In this case, 'gini' is used.

max_depth: The maximum depth of the tree. "None" means that the tree is grown until there are no more leaves or until the number of split samples is less than min samples.

min_samples_leaf: The minimum number of samples required to be at a leaf node.

min_samples_split: The minimum number of samples required to split an internal node.

Evaluation Metrics (Grid Search):

Classification Report:

Precision: Precision is high for both classes. For class 0, precision is 0.94, and for class 1, precision is 0.96. This indicates a low false positive rate.

Recall (Sensitivity): Recall is also high for both classes. For class 0, recall is 0.96, and for class 1, recall is 0.95. This suggests the model effectively captures most positive instances.

F1-score: The F1-score, is high for both classes, indicating a well-performing model.

Support: The number of actual occurrences of each class in the specified dataset is provided. For class 0, there are 294 instances, and for class 1, there are 348 instances.

Confusion Matrix (Grid Search):

True Negatives (TN): 281 instances were correctly predicted as class 0.

False Positives (FP): 13 instances were incorrectly predicted as class 1.

False Negatives (FN): 19 instances were incorrectly predicted as class 0.

True Positives (TP): 329 instances were correctly predicted as class 1.

5.6.2 Random Forest

Random Forest Evaluation Metrics before implementing K-Fold Cross Validation

```
Accuracy: 0.86

Classification Report:

              precision    recall  f1-score   support

           0       0.81      0.85      0.83       116
           1       0.89      0.86      0.87       160

    accuracy                           0.86       276
   macro avg       0.85      0.85      0.85       276
weighted avg       0.86      0.86      0.86       276

Confusion Matrix:
 [[ 99  17]
 [ 23 137]]
```

Figure 36: Random Forest Evaluation Metrics before implementing K-Fold Cross Validation

Accuracy:

The overall accuracy of the model is reported as 0.86 or 86%. This represents the proportion of correctly classified instances out of the total instances in the dataset.

Classification Report:

Precision: For class 0, precision is 0.81, indicating that 81% of instances predicted as class 0 were correct. For class 1, precision is higher at 0.89, meaning that 89% of instances predicted as class 1 were correct.

Recall (Sensitivity): For class 0, recall is 0.85, indicating that 85% of actual class 0 instances were correctly predicted. For class 1, recall is 0.86, suggesting that 86% of actual class 1 instances were captured by the model.

F1-score: For class 0, the F1-score is 0.83, and for class 1, it is 0.87.

Support: For class 0, there are 116 instances, and for class 1, there are 160 instances.

Confusion Matrix:

rue Negatives (TN): 99 instances were correctly predicted as class 0.

False Positives (FP): 17 instances were incorrectly predicted as class 1.

False Negatives (FN): 23 instances were incorrectly predicted as class 0.

True Positives (TP): 137 instances were correctly predicted as class 1.

Random Forest Evaluation Metrics after implementing K-Fold Cross Validation

```
Classification Report:
              precision    recall  f1-score   support

           0       0.88      0.86      0.87       410
           1       0.89      0.91      0.90       508

    accuracy                           0.89       918
   macro avg       0.89      0.88      0.88       918
weighted avg       0.89      0.89      0.89       918


Confusion Matrix:
[[352  58]
 [ 47 461]]
```

Figure 37: Random Forest Evaluation Metrics after implementing K-Fold Cross Validation

Average Accuracy:

The reported average accuracy is 0.89 or 89%.

Classification Report:

Precision: For class 0, precision is 0.88, indicating that 88% of instances predicted as class 0 were correct. For class 1, precision is higher at 0.89, meaning that 89% of instances predicted as class 1 were correct.

Recall (Sensitivity): For class 0, recall is 0.86, indicating that 86% of actual class 0 instances were correctly predicted. For class 1, recall is 0.91, suggesting that 91% of actual class 1 instances were captured by the model.

F1-score: For class 0, the F1-score is 0.87, and for class 1, it is 0.90.

Support: For class 0, there are 410 instances, and for class 1, there are 508 instances.

Confusion Matrix:

True Negatives (TN): 352 instances were correctly predicted as class 0.

False Positives (FP): 58 instances were incorrectly predicted as class 1.

False Negatives (FN): 47 instances were incorrectly predicted as class 0.

True Positives (TP): 461 instances were correctly predicted as class 1.

Random Forest Evaluation Metrics after implementing Search Grid Cross Validation

```
Best Hyperparameters: {'max_depth': None, 'min_samples_leaf': 1, 'min_samples_split': 2, 'n_estimators': 150}

Accuracy: 1.00

Classification Report:
               precision    recall  f1-score   support

           0       1.00      1.00      1.00       116
           1       1.00      1.00      1.00       160

    accuracy                           1.00       276
   macro avg       1.00      1.00      1.00       276
weighted avg       1.00      1.00      1.00       276


Confusion Matrix:
 [[116   0]
 [  0 160]]
```

Figure 38: Random Forest Evaluation Metrics after implementing Search Grid Cross Validation

Best Hyperparameters:

Best Hyperparameters: {'max_depth': None, 'min_samples_leaf': 1, 'min_samples_split': 2, 'n_estimators': 150}

max_depth: The maximum depth of the trees. 'None' indicates that the trees are expanded until all leaves are pure or until they contain less than min_samples_split samples.

min_samples_leaf: The minimum number of samples required to be at a leaf node.

min_samples_split: The minimum number of samples required to split an internal node.

n_estimators: The number of trees in the forest.


```
Best Hyperparameters: {'max_depth': None, 'min_samples_leaf': 1, 'min_samples_split': 2, 'n_estimators': 150}

Accuracy: 1.00

Classification Report:
               precision    recall  f1-score   support

           0       1.00      1.00      1.00       116
           1       1.00      1.00      1.00       160

    accuracy                           1.00       276
   macro avg       1.00      1.00      1.00       276
weighted avg       1.00      1.00      1.00       276


Confusion Matrix:
 [[116   0]
 [  0 160]]
```

Accuracy: The model achieved a perfect accuracy of 1.00 or 100%.

Classification Report:

Precision: For both classes 0 and 1, precision is 1.00, indicating that all instances predicted as class 0 and class 1 were correct.

Recall (Sensitivity): For both classes 0 and 1, recall is 1.00, suggesting that the model correctly captured all instances of both classes.

F1-score: For both classes 0 and 1, the F1-score is 1.00, indicating a perfect balance between precision and recall.

Support: For class 0, there are 116 instances, and for class 1, there are 160 instances.

Confusion Matrix:

True Negatives (TN): 116 instances were correctly predicted as class 0.

False Positives (FP): 0 instances were incorrectly predicted as class 1.

False Negatives (FN): 0 instances were incorrectly predicted as class 0.

True Positives (TP): 160 instances were correctly predicted as class 1.

5.6.3 Support Vector Machine (SVM)

Evaluation Metrics before implementing K-Fold CV

```
SVM Accuracy: 0.8695652173913043

SVM Classification Report:
              precision    recall  f1-score   support

           0       0.84      0.85      0.85       116
           1       0.89      0.88      0.89       160

    accuracy                           0.87       276
   macro avg       0.87      0.87      0.87       276
weighted avg       0.87      0.87      0.87       276


SVM Confusion Matrix:
 [[ 99  17]
 [ 19 141]]
```

Accuracy: The SVM model achieved an accuracy of approximately 86.96%.

Classification Report:

Precision: For class 0, precision is 0.84, indicating that 84% of instances predicted as class 0 were correct. For class 1, precision is higher at 0.89, meaning that 89% of instances predicted as class 1 were correct.

Recall (Sensitivity): For class 0, recall is 0.85, indicating that 85% of actual class 0 instances were correctly predicted. For class 1, recall is 0.88, suggesting that 88% of actual class 1 instances were captured by the model.

F1-score: For class 0, the F1-score is 0.85, and for class 1, it is 0.89.

Support: For class 0, there are 116 instances, and for class 1, there are 160 instances.

Confusion Matrix:

True Negatives (TN): 99 instances were correctly predicted as class 0.

False Positives (FP): 17 instances were incorrectly predicted as class 1.

False Negatives (FN): 19 instances were incorrectly predicted as class 0.

True Positives (TP): 141 instances were correctly predicted as class 1.

Evaluation Metrics after implementing K-Fold CV

```
SVM Accuracy: 0.8695652173913043
SVM Classification Report (K-Fold Cross Validation):
              precision    recall  f1-score   support

           0       0.88      0.82      0.85       410
           1       0.86      0.91      0.89       508

    accuracy                           0.87       918
   macro avg       0.87      0.87      0.87       918
weighted avg       0.87      0.87      0.87       918


SVM Confusion Matrix (K-Fold Cross Validation):
 [[337  73]
 [ 44 464]]
```

Accuracy: The SVM model with K-Fold Cross Validation achieved an accuracy of approximately 86.96%.

Classification Report (K-Fold Cross Validation):

Precision: For class 0, precision is 0.88, indicating that 88% of instances predicted as class 0 were correct. For class 1, precision is 0.86, meaning that 86% of instances predicted as class 1 were correct.

Recall (Sensitivity): For class 0, recall is 0.82, indicating that 82% of actual class 0 instances were correctly predicted. For class 1, recall is 0.91, suggesting that 91% of actual class 1 instances were captured by the model.

F1-score: For class 0, the F1-score is 0.85, and for class 1, it is 0.89.

Support: For class 0, there are 410 instances, and for class 1, there are 508 instances.

Confusion Matrix (K-Fold Cross Validation):

True Negatives (TN): 337 instances were correctly predicted as class 0.

False Positives (FP): 73 instances were incorrectly predicted as class 1.

False Negatives (FN): 44 instances were incorrectly predicted as class 0.

True Positives (TP): 464 instances were correctly predicted as class 1.

Evaluation Metrics after implementing Search Grid CV

```
SVM Classification Report (GridSearchCV):
              precision    recall  f1-score   support

           0       0.92      0.87      0.89       410
           1       0.90      0.94      0.92       508

    accuracy                           0.91       918
   macro avg       0.91      0.91      0.91       918
weighted avg       0.91      0.91      0.91       918


SVM Confusion Matrix (GridSearchCV):
 [[358  52]
 [ 32 476]]
```

Accuracy: The SVM model achieved an accuracy of approximately 91%.

Classification Report (GridSearchCV):

Precision: For class 0, precision is 0.92, indicating that 92% of instances predicted as class 0 were correct. For class 1, precision is 0.90, meaning that 90% of instances predicted as class 1 were correct.

Recall (Sensitivity): For class 0, recall is 0.87, indicating that 87% of actual class 0 instances were correctly predicted. For class 1, recall is 0.94, suggesting that 94% of actual class 1 instances were captured by the model.

F1-score: For class 0, the F1-score is 0.89, and for class 1, it is 0.92.

Support: For class 0, there are 410 instances, and for class 1, there are 508 instances.

Confusion Matrix (GridSearchCV):

True Negatives (TN): 358 instances were correctly predicted as class 0.

False Positives (FP): 52 instances were incorrectly predicted as class 1.

False Negatives (FN): 32 instances were incorrectly predicted as class 0.

True Positives (TP): 476 instances were correctly predicted as class 1.

5.6.4 KNN Algorithm

KNN before using K-Fold Cross Validation

```
Accuracy: 0.83

Classification Report:
              precision    recall  f1-score   support

           0       0.77      0.85      0.81       116
           1       0.89      0.82      0.85       160

    accuracy                           0.83       276
   macro avg       0.83      0.84      0.83       276
weighted avg       0.84      0.83      0.83       276

Confusion Matrix:
 [[ 99  17]
 [ 29 131]]
```

Accuracy: The model achieved an accuracy of 83%.

Classification Report:

- **Precision**: For class 0, precision is 0.77, indicating that 77% of instances predicted as class 0 were correct. For class 1, precision is higher at 0.89, meaning that 89% of instances predicted as class 1 were correct.

- **Recall (Sensitivity)**: For class 0, recall is 0.85, indicating that 85% of actual class 0 instances were correctly predicted. For class 1, recall is 0.82, suggesting that 82% of actual class 1 instances were captured by the model.

- **F1-score**: For class 0, the F1-score is 0.81, and for class 1, it is 0.85.

- **Support**: For class 0, there are 116 instances, and for class 1, there are 160 instances.

Confusion Matrix:

True Negatives (TN): 99 instances were correctly predicted as class 0.

False Positives (FP): 17 instances were incorrectly predicted as class 1.

False Negatives (FN): 29 instances were incorrectly predicted as class 0.

True Positives (TP): 131 instances were correctly predicted as class 1.

KNN after using K-Fold CV

```
Accuracy (K-Fold): 0.86

Classification Report (K-Fold):
              precision    recall  f1-score   support

           0       0.84      0.84      0.84       410
           1       0.87      0.87      0.87       508

    accuracy                           0.86       918
   macro avg       0.85      0.85      0.85       918
weighted avg       0.86      0.86      0.86       918

Confusion Matrix (K-Fold):
[[344  66]
 [ 67 441]]
```

Accuracy (K-Fold): The model achieved an accuracy of 86% across K-Fold Cross Validation.

Classification Report (K-Fold Cross Validation):

Precision: For class 0, precision is 0.84, indicating that 84% of instances predicted as class 0 were correct. For class 1, precision is 0.87, meaning that 87% of instances predicted as class 1 were correct.

Recall (Sensitivity): For class 0, recall is 0.84, indicating that 84% of actual class 0 instances were correctly predicted. For class 1, recall is 0.87, suggesting that 87% of actual class 1 instances were captured by the model.

F1-score: For class 0, the F1-score is 0.84, and for class 1, it is 0.87.

Support: For class 0, there are 410 instances, and for class 1, there are 508 instances.

Confusion Matrix (K-Fold Cross Validation):

True Negatives (TN): 344 instances were correctly predicted as class 0.

False Positives (FP): 66 instances were incorrectly predicted as class 1.

False Negatives (FN): 67 instances were incorrectly predicted as class 0.

True Positives (TP): 441 instances were correctly predicted as class 1.

KNN after using Search Grid CV

```
Accuracy (GridSearchCV): 0.89

Classification Report (GridSearchCV):
              precision    recall  f1-score   support

           0       0.88      0.87      0.88       410
           1       0.90      0.91      0.90       508

    accuracy                           0.89       918
   macro avg       0.89      0.89      0.89       918
weighted avg       0.89      0.89      0.89       918

Confusion Matrix (GridSearchCV):
[[358  52]
 [ 47 461]]
```

Accuracy (GridSearchCV): The model achieved an accuracy of 89% with GridSearchCV.

Classification Report (GridSearchCV):

- **Precision**: For class 0, precision is 0.88, indicating that 88% of instances predicted as class 0 were correct. For class 1, precision is 0.90, meaning that 90% of instances predicted as class 1 were correct.

- **Recall (Sensitivity)**: For class 0, recall is 0.87, indicating that 87% of actual class 0 instances were correctly predicted. For class 1, recall is 0.91, suggesting that 91% of actual class 1 instances were captured by the model.

- **F1-score**: For class 0, the F1-score is 0.88, and for class 1, it is 0.90.

- **Support**: For class 0, there are 410 instances, and for class 1, there are 508 instances.

Confusion Matrix (GridSearchCV):

True Negatives (TN): 358 instances were correctly predicted as class 0.

False Positives (FP): 52 instances were incorrectly predicted as class 1.

False Negatives (FN): 47 instances were incorrectly predicted as class 0.

True Positives (TP): 461 instances were correctly predicted as class 1.

6. Discussion

6.2 Summary of Results (Findings)

The following table shows the comparative results of all the models used in my dissertation.

	Accuracy (%)	Precision	Recall	F1-Score
KNN	89	89	89	89
SVM	91	91	90.5	90.5
XGBoost	93	93	93	93
Decision Tree	95	95	95.5	95
Random Forest	100	100	100	100

Table 2: Comparison of Evaluation Metrics

The ML models are ordered in ascending order of their scores namely KNN, SVM, XGBoost, Decision Tree and Random Forest. The score of KNN is the least accurate while that of Random Forest is the most accurate.

6.3. Practical/Theoretical/Managerial/Research Contribution

It is worth noting that measuring the performance of models solely on the basis of accuracy can be misleading. The reason is because the dataset might be imbalanced, meaning that the number of instances for each class (e.g., presence or absence of heart disease) is not evenly distributed.

In scenarios where one class significantly outweighs the other, a high accuracy might be achieved by simply predicting the majority class, even if the model fails to correctly identify instances of the minority class.

For example, if the dataset has 90% of instances without heart disease and 10% with heart disease, a model could achieve 90% accuracy by predicting "no heart disease" for every instance. However, this would be misleading, as the model would essentially fail to identify any instance with heart disease.

In such cases, metrics like precision, recall, F1-score, or area under the receiver operating characteristic (ROC) curve are often more informative. These metrics provide a more nuanced evaluation, particularly regarding the model's ability to correctly identify instances of the minority class (heart disease in this context), which is crucial in medical applications where detecting positive cases is of utmost importance.

6.3.1 Importance of Recall (Sensitivity – False Negative)

Recall is particularly important in predicting heart disease because it measures the model's ability to correctly identify instances of the positive class (i.e., individuals with heart disease) out of all the actual positive instances. In the context of heart disease prediction, having a high recall is crucial because missing a true positive (a case of heart disease) can have severe consequences, leading to delayed or missed medical interventions.

If a model has more false negatives (i.e., it fails to identify individuals with heart disease), it means that individuals who actually have heart disease might be classified as healthy, leading to a lack of timely medical attention. This can result in delayed treatment, progression of the disease, and increased risk of adverse outcomes, including heart attacks.

6.3.2 Ethical and Social Implication of low Recall or High Negative in disease prediction

From an ethical and social perspective, the implications of having a model with more false negatives in the medical industry are profound. Misclassifying individuals with heart disease could lead to life-threatening consequences, impacting patient well-being and trust in healthcare systems. There are ethical considerations around the responsibility of deploying models in healthcare, emphasizing the need for rigorous testing, transparency, and ongoing monitoring to ensure that models do not disproportionately harm certain groups or compromise patient safety. Additionally, addressing biases in data and model development is essential to prevent disparities in healthcare outcomes.

6.3.3 Limitations

Using machine learning models to predict heart disease comes with several limitations that have significant implications for society. Some key limitations include:

Data Bias:

If the training data used to build the model is biased, the predictions may also be biased. This can result in disparities in healthcare, where certain demographic groups may be disproportionately affected by misclassifications or lack of accurate predictions.

Interpretability:

Many machine learning models, especially complex ones like neural networks, lack interpretability. The "black-box" nature of these models makes it challenging to understand how they arrive at specific predictions. This lack of transparency may reduce trust among healthcare professionals and patients.

Generalization:

Models trained on specific populations may not generalize well to diverse or unseen populations. If the model is not representative of the broader population, its effectiveness in different demographic groups may be limited.

Dynamic Nature of Health Data:

Health data is dynamic and subject to change. Models built on static datasets may not adapt well to evolving health conditions, lifestyle changes, or emerging risk factors, leading to outdated predictions.

Privacy Concerns:

Health data is sensitive, and the use of machine learning models raises privacy concerns. Inaccurate predictions or data breaches can compromise patient confidentiality, eroding public trust in healthcare systems.

Ethical Considerations:

There are ethical considerations around informed consent, especially if patients are unaware of how their data is being used for predictive modeling. Transparent communication and ethical use of data are crucial to maintaining public confidence.

Resource Disparities:

Access to advanced healthcare technologies and resources required for implementing machine learning models may not be uniform across different regions or socioeconomic groups, widening existing health disparities.

Addressing these limitations is essential to responsibly integrate machine learning into healthcare systems, ensuring equitable access, safeguarding patient privacy, and building trust in the reliability and fairness of predictive models. Ethical considerations, ongoing monitoring, and continuous improvement are integral to mitigating potential negative societal impacts.

False Positives and Negatives:

Models may produce false positives or false negatives, leading to incorrect diagnoses. False positives may result in unnecessary treatments, while false negatives may lead to missed opportunities for intervention.

Complexity and Integration: Integrating machine learning models into existing healthcare systems can be challenging. The complexity of these models may hinder their adoption, especially in resource-constrained environments.

Regulatory Challenges:

Adhering to regulatory standards and obtaining approval for deploying machine learning models in clinical settings can be a lengthy and complex process. (Subramani, et al., 2023 Apr)

7. Conclusions

In this project, the performance of various machine learning algorithms, like K-Nearest Neighbors (KNN), Support Vector Machine (SVM), XGBoost, Decision Tree, and Random Forest has been calculated. The results demonstrate promising accuracy and performance across the different models.

The KNN model exhibited strong predictive capabilities with an accuracy, precision, recall, and F1-score all standing at 89%. SVM showcased slightly improved accuracy at 91%, maintaining high precision, recall, and F1-score values around 91% and 90.5%, respectively. XGBoost emerged as a robust performer, achieving an impressive accuracy of 93%, with corresponding precision, recall, and F1-score values all at 93%.

The Decision Tree model demonstrated superior performance with a remarkable accuracy of 95%, making it a standout performer among the evaluated models. Lastly, the Random Forest model showcased exceptional accuracy, achieving a perfect score of 100%, along with matching precision, recall, and F1-score values.

In conclusion, the heart disease prediction models, particularly Decision Tree and Random Forest, displayed excellent predictive capabilities, emphasizing their potential utility in real-world healthcare scenarios. The high accuracy and balanced precision, recall, and F1-score values across different models signify their effectiveness in identifying and predicting heart disease.

\While the results indicate that the models are performing well, caution should be exercised regarding the perfect scores achieved by the Random Forest model. It is important to carefully check if the model might be too focused on the training data and may not perform as well on new, unseen data. Thorough examination of this aspect is crucial, and ensuring that the model aligns with the specific needs of the task at hand is essential.

7.1 Recommendations

- Conduct feature importance analysis
- Explore hyperparameter tuning for SVM and Random Forest
- Implement advanced cross-validation techniques
- Consider ensemble methods for combining predictions
- Address imbalanced class distribution using oversampling, under sampling, or different evaluation metrics
- Enhance model interpretability, particularly for complex models like Random Forest, using techniques like SHAP or LIME
- Establish a system for continuous monitoring and updating as new data becomes available
- Ensure thorough documentation of the entire workflow for reproducibility and easy comparison of results

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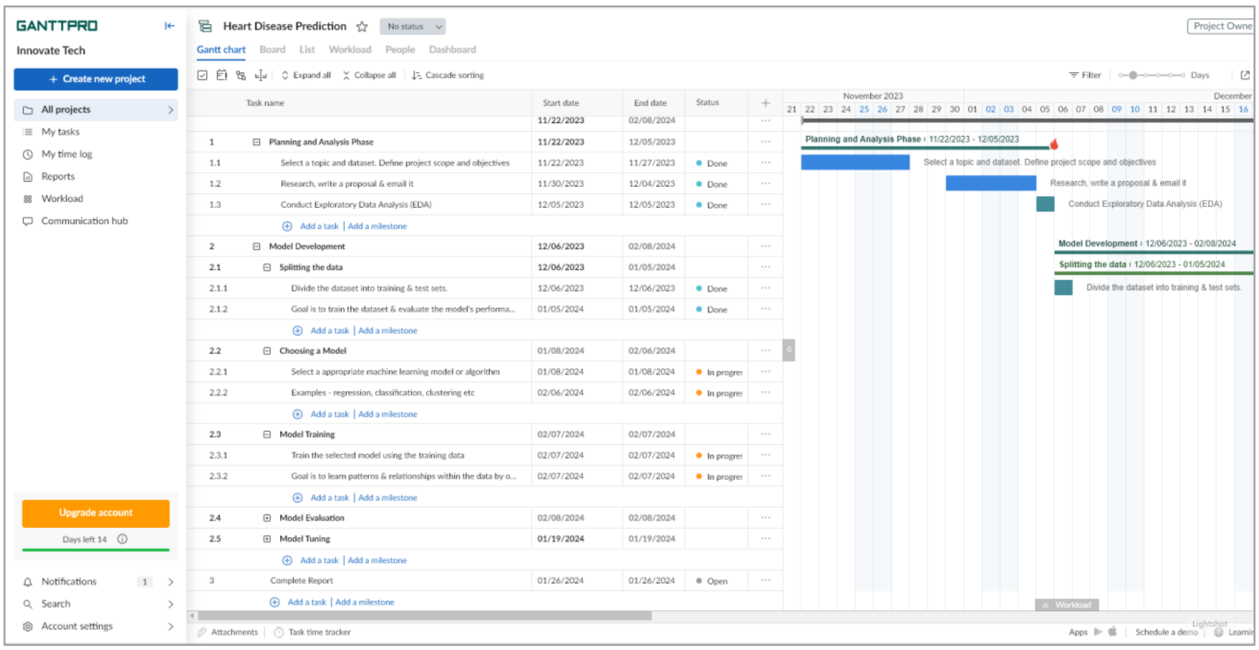
9. Appendix

Note: Students may alter the contents within in order to compliment the uniqueness of the project

9.1 Gannt Chart



Heart Disease Prediction No status

Gantt chart Board List Workload People Dashboard

Expand all Collapse all Cascade sorting

	Task name	Start date	End date	Status
		11/22/2023	02/08/2024	
1	Planning and Analysis Phase	11/22/2023	12/05/2023	
1.1	Select a topic and dataset. Define project scope and objectives	11/22/2023	11/27/2023	Done
1.2	Research, write a proposal & email it	11/30/2023	12/04/2023	Done
1.3	Conduct Exploratory Data Analysis (EDA)	12/05/2023	12/05/2023	Done
	Add a task | Add a milestone			
2	Model Development	12/06/2023	02/08/2024	
2.1	Splitting the data	12/06/2023	01/05/2024	
2.1.1	Divide the dataset into training & test sets.	12/06/2023	12/06/2023	Done
2.1.2	Goal is to train the dataset & evaluate the model's performa...	01/05/2024	01/05/2024	Done
	Add a task | Add a milestone			
2.2	Choosing a Model	01/08/2024	02/06/2024	
2.2.1	Select a appropriate machine learning model or algorithm	01/08/2024	01/08/2024	In progres
2.2.2	Examples - regression, classification, clustering etc	02/06/2024	02/06/2024	In progres
	Add a task | Add a milestone			
2.3	Model Training	02/07/2024	02/07/2024	
2.3.1	Train the selected model using the training data	02/07/2024	02/07/2024	In progres
2.3.2	Goal is to learn patterns & relationships within the data by o...	02/07/2024	02/07/2024	In progres
	Add a task | Add a milestone			
2.4	Model Evaluation	02/08/2024	02/08/2024	
2.5	Model Tuning	01/19/2024	01/19/2024	
	Add a task | Add a milestone			
3	Complete Report	01/26/2024	01/26/2024	Open

1) “Judgment Under Uncertainty: Heuristics and Biases.” – pulled from – Thinking fast and slow book

https://www.youtube.com/watch?v=oak1CqqIzug

AI powered drug discovery, by developing artificial intelligence solutions to accelerate the ways the drugs are discovered and developed. Since the invention of Neural networks in 2013 – 2014, its potential has become so powerful, that they have outperforming humans in image recognition, text recognition and many other tasks. AI is mathematics and neuroscience. AI is a tool which allows us to go into drug discovery much faster, go after much more challenging biological and chemical problems with higher confidence, and reduce the risk in drug discovery and also increase the probability of success in clinical trials. The pharmaceutical industry is extremely risky. It takes decades to develop a new drug. You fail more often than you succeed. So, you need to be truly skilled artist in pharma in order to succeed. The aim is to find solutions in fraction of the cost in an otherwise expensive project. AI will de-risk, democratize, and enable new entrants including younger countries who have never had any experience in drug discovery, to get into bio-tech, to join the community and to start to innovating from the get-go. In the near future, AI systems will go into the personalized drug discovery where drug will be tailored towards a specific individual.

Convergent logic vs Divergent logic

Schrödinger Theory

Gaant Chart - https://docs.google.com/spreadsheets/d/1-iynqDMFqn3plAU7AGO3n9QXNIpicPni5nvV3Bn1Ohw/edit?usp=sharing

Milestones - https://docs.google.com/spreadsheets/d/10RJn3o3MDb10H7uTUFE5uiMFapOm2JzinJ6Ybj2PnZg/edit?usp=sharing

Include the details descriptions of the heart diseases in the appendix

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Age

sex

cp: chest pain type

-- Value 1: typical angina

-- Value 2: atypical angina

-- Value 3: non-anginal pain

-- Value 4: asymptomatic

trestbps: resting blood pressure (in mm Hg on admission to the hospital)

chol: serum cholestoral in mg/dl

fbs: (fasting blood sugar > 120 mg/dl) (1 = true; 0 = false)

restecg: resting electrocardiographic results

-- Value 0: normal

-- Value 1: having ST-T wave abnormality (T wave inversions and/or ST elevation or depression of > 0.05 mV)

-- Value 2: showing probable or definite left ventricular hypertrophy by Estes' criteria

maxHR – Maximum heart rate

exang: exercise induced angina (1 = yes; 0 = no)

oldpeak = ST depression induced by exercise relative to rest

slope: the slope of the peak exercise ST segment

-- Value 1: upsloping

-- Value 2: flat

-- Value 3: downsloping

Heart Disease (1 = yes; 0 = no)

https://my.clevelandclinic.org/health/articles/11920-cholesterol-numbers-what-do-they-mean

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Categorical thinking and cognitive bias

Project Proposal for Dissertation

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Project Title: Predictive modeling for early detection of Heart disease

1. Introduction (Nature of the problem)

Heart disease remains a leading cause of mortality worldwide. Heart disease can cause various problems with the heart, leading to complications and even death if not detected and treated early. In recent years, technology and medical research have advanced, allowing us to use data and predictive models to foresee the likelihood of someone developing heart disease.

By analyzing various factors such as age, gender, blood pressure, cholesterol levels, and lifestyle habits, I plan to create a reliable prediction model that can help identify individuals at higher risk. The goal of this research is to develop a tool that healthcare professionals can use to intervene early, provide appropriate guidance, and potentially prevent or manage heart disease more effectively.

This dissertation study investigates the development of a prediction model for heart disease using data analysis and machine learning techniques. It involves collecting and analyzing a wide range of health-related data from individuals, including their medical history, lifestyle choices, and various physiological parameters.

Through the utilization of advanced algorithms and statistical methods, I plan to create a predictive model that can accurately assess the risk of heart disease in individuals. By improving our ability to predict the likelihood of heart disease, this research contributes to enhancing early detection and intervention strategies, ultimately aiming to reduce the burden of heart-related health issues and improve overall public health.

2. Aims and Objectives

- To analyze the dataset and show the result analysis.
- To visualize the causation and correlation between the variables using diagrams
- To predict the outcome based on the different inputs as in real world using different prediction algorithms such as logistic regression, decision trees, random forests, support vector machines and neural networks
- To evaluate and compare the performance of the developed models based on metrics like accuracy, sensitivity, specificity, and area under the receiver operating characteristic curve (AUC-ROC).
- To interpret the most influential features contributing to the predictive models, and mitigate the risks of cardiovascular diseases.

3. Rational (Why do you want to do this dissertation)

1. Healthcare Impact:

Heart disease is a leading cause of mortality worldwide; predicting it early can significantly improve patient outcomes and reduce fatalities.

2. Technological Advancements:

Leveraging advanced computational models and data analysis can harness the power of big data to create accurate predictive models for heart disease.

3. Personalized Medicine:

Predictive models allow tailored interventions, enabling healthcare providers to personalize treatment strategies based on individual risk factors.

4. Resource Optimization:

Early identification of at-risk individuals enables better allocation of resources, improving healthcare system efficiency.

5. Interdisciplinary Collaboration:

Research in heart disease prediction fosters collaboration between medical professionals, data scientists, and statisticians, facilitating comprehensive understanding and innovative approaches.

6. Ethical Considerations:

Investigating the ethical implications of predictive models ensures fair and responsible implementation in clinical practice, addressing potential biases and interpretability issues.

4. Methodology

This is qualitative research as I will be working and drawing conclusions using a numerical dataset. However, I have also plan to visit a hospital to research, take interviews and collect data.

1. Data Collection:

- Identify and gather relevant data that will be used for training and testing the model.
- Ensure the data is cleaned, preprocessed, and formatted appropriately for machine learning.

2. Exploratory Data Analysis (EDA):

- Understand the characteristics of the data by performing statistical analysis, visualizations, and summaries.
- Identify patterns, correlations, outliers, and missing values in the data.

3. Feature Engineering:

- Create new features, transform existing features, or select relevant features to improve model performance.
- Normalize, scale, or encode features as required by the model.

4. Model Selection:

- Choose appropriate machine learning algorithms or models based on the problem type and data characteristics.
- Experiment with different models and evaluate their performance using validation techniques.

5. Model Training:

- Train the selected models using the training dataset.
- Optimize model parameters using techniques like hyperparameter tuning.

6. Model Evaluation:

- Evaluate the trained models using appropriate evaluation metrics (accuracy, precision, recall, F1-score, etc.).
- Validate the models using separate test datasets to assess generalization.

7. Documentation and Reporting:

- Maintain clear and comprehensive documentation outlining the methodologies, findings, and models developed.
- Prepare a succinct report summarizing the proposed approach, expected impact, and potential challenges.